

CHAPTER 17

ELECTRICITY

4.2 Electrical quantities

4.2.1 Electrical charge

- 1 State that there are positive and negative charges and that charge is measured in coulombs
- 2 State that unlike charges attract and like charges repel
- 3 Describe experiments to show electrostatic charging by friction
- 4 Explain that charging of solids by friction involves only a transfer of negative charge (electrons)
- 5 Describe an electric field as a region in which an electric charge experiences a force
- 6 State that the direction of an electric field line at a point is the direction of the force on a positive charge at that point
- 7 Describe simple electric field patterns, including the direction of the field:
 - (a) around a point charge
 - (b) around a charged conducting sphere
 - (c) between two oppositely charged parallel conducting plates (end effects will **not** be examined)
- 8 State examples of electrical conductors and insulators
- 9 Describe an experiment to distinguish between electrical conductors and insulators
- 10 Recall and use a simple electron model to explain the difference between electrical conductors and insulators

4.2.2 Electrical current

- 1 Define electric current as the charge passing a point per unit time; recall and use the equation
$$\text{electric current} = \frac{\text{charge}}{\text{time}}$$
$$I = \frac{Q}{t}$$
- 2 Describe electrical conduction in metals in terms of the movement of free electrons
- 3 Know that current is measured in amps (amperes) and that the amp is given by coulomb per second (C/s)
- 4 Know the difference between direct current (d.c.) and alternating current (a.c.)
- 5 State that conventional current is from positive to negative and that the flow of free electrons is from negative to positive
- 6 Describe the use of ammeters (analogue and digital) with different ranges

4.2.3 Electromotive force and potential difference

- 1 Define e.m.f. (electromotive force) as the electrical work done by a source in moving a unit charge around a complete circuit; recall and use the equation
$$\text{e.m.f.} = \frac{\text{work done (by a source)}}{\text{charge}}$$

$$E = \frac{W}{Q}$$

continued

4.2.3 Electromotive force and potential difference continued

- 2 Define p.d. (potential difference) as the work done by a unit charge passing through a component; recall and use the equation

$$\text{p.d.} = \frac{\text{work done (on a component)}}{\text{charge}}$$

$$V = \frac{W}{Q}$$

- 3 Know that e.m.f. and p.d. are measured in volts and that the volt is given by joule per coulomb (J/C)
- 4 Describe the use of voltmeters (analogue and digital) with different ranges
- 5 Calculate the total e.m.f. where several sources are arranged in series
- 6 State that the e.m.f. of identical sources connected in parallel is equal to the e.m.f. of one of the sources

4.2.4 Resistance

- 1 Recall and use the equation

$$\text{resistance} = \frac{\text{p.d.}}{\text{current}}$$

$$R = \frac{V}{I}$$

- 2 Describe an experiment to determine resistance using a voltmeter and an ammeter and do the appropriate calculations
- 3 Recall and use, for a wire, the direct proportionality between resistance and length, and the inverse proportionality between resistance and cross-sectional area
- 4 State Ohm's law, including reference to constant temperature
- 5 Sketch and explain the current-voltage graphs for a resistor of constant resistance, a filament lamp and a diode
- 6 Describe the effect of temperature increase on the resistance of a resistor, such as the filament in a filament lamp

4.3 Electric circuits

4.3.1 Circuit diagrams and circuit components

- 1 Draw and interpret circuit diagrams with cells, batteries, power supplies, generators, oscilloscopes, potential dividers, switches, resistors (fixed and variable), heaters, thermistors (NTC only), light-dependent resistors (LDRs), lamps, motors, ammeters, voltmeters, magnetising coils, transformers, fuses, relays, diodes and light-emitting diodes (LEDs), and know how these components behave in the circuit

4.3.2 Series and parallel circuits

- 1 Recall and use in calculations, the fact that:
 - (a) the current at every point in a series circuit is the same
 - (b) the sum of the currents entering a junction in a parallel circuit is equal to the sum of the currents that leave the junction
 - (c) the total p.d. across the components in a series circuit is equal to the sum of the individual p.d.s across each component
 - (d) the p.d. across an arrangement of parallel resistances is the same as the p.d. across one branch in the arrangement of the parallel resistances
- 2 Calculate the combined resistance of two or more resistors in series
- 3 Calculate the combined resistance of two resistors in parallel
- 4 Calculate current, voltage and resistance in parts of a circuit or in the whole circuit

4.3.3 Action and use of circuit components

- 1 Describe the action of negative temperature coefficient (NTC) thermistors and light-dependent resistors and explain their use as input sensors
- 2 Describe the action of a variable potential divider
- 3 Recall and use the equation for two resistors used as a potential divider

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$

1. O/N/23/11/Q23

An aircraft flying through the air becomes electrically charged.

What causes this to happen?

- A** background radiation in the aircraft
- B** friction of the aircraft with the air
- C** heating of the aircraft surface
- D** radio waves emitted by the aircraft

2. O/N/23/11/Q24

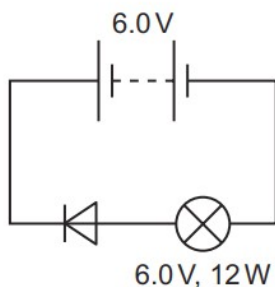
Two metal spheres are brought close to each other. The two spheres repel each other.

Which statement explains this?

- A** Both spheres are charged and both are positive.
- B** Both spheres are charged but one is positive and the other is negative.
- C** The spheres have the same number of electrons.
- D** The spheres have the same number of protons.

3. O/N/23/11/Q25

A student sets up the circuit shown. The lamp does not light.

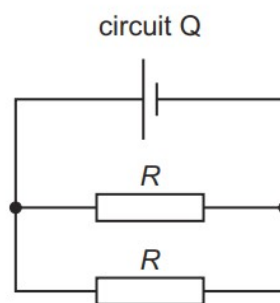
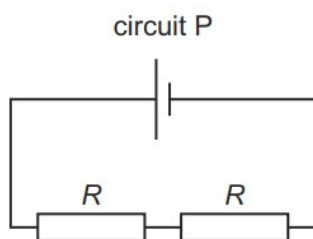


What makes the lamp light?

- A** changing the supply to a 12 V battery
- B** inserting a fuse
- C** reversing the battery connections
- D** reversing the connections to the lamp

4. O/N/23/11/Q26

Circuits P and Q each contain two identical resistors, each of resistance R .



What is the resistance of circuit P and what is the resistance of circuit Q?

	resistance of circuit P	resistance of circuit Q
A	R	$\frac{R}{2}$
B	R	$2R$
C	$2R$	$\frac{R}{2}$
D	$2R$	$2R$

5. O/N/23/11/Q27

The current in an electric motor is 2.0 A.

It transfers 20 J of energy in 5.0 s.

What is the potential difference (p.d.) across the motor?

- A** 0.50 V **B** 2.0 V **C** 8.0 V **D** 50 V

6. O/N/23/12/Q31

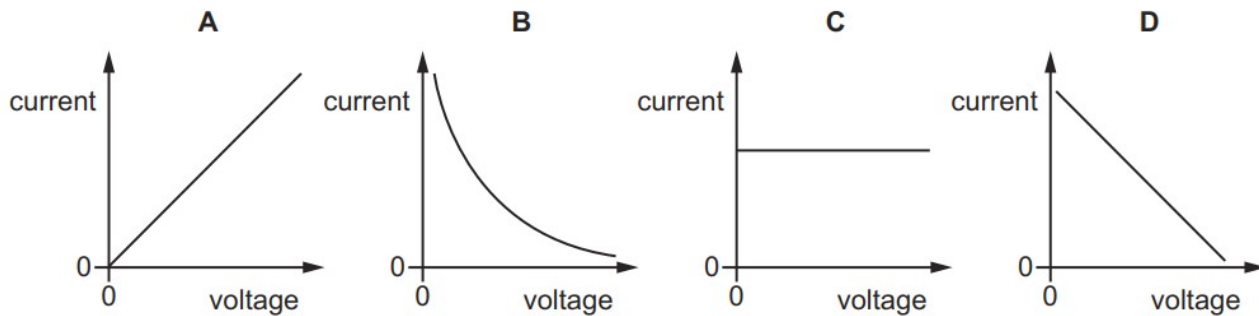
A student rubs different rods held in his hand.

Which rod gains a large electrostatic charge?

- A** a polythene rod rubbed with a steel magnet
B a polythene rod rubbed with a woollen duster
C an iron rod rubbed with a steel magnet
D an iron rod rubbed with a woollen duster

7. O/N/23/12/Q33

Which graph shows how the current changes when the voltage across a resistor of constant resistance is varied?

**8. O/N/23/12/Q34**

Two identical resistors are connected first in series and then in parallel.

The two resistors in series are connected to a power supply and the current is I_s . The parallel combination is connected to the same power supply and the current supplied is I_p .

What is the ratio $\frac{I_s}{I_p}$?

- A** 0.25 **B** 0.50 **C** 2.0 **D** 4.0

9. M/J/23/12/Q26

A shoe becomes positively charged by friction when it rubs against a carpet.

What happens as the shoe becomes charged?

- A** Negative electrons are transferred to the carpet.
- B** Negative electrons are transferred to the shoe.
- C** Positive electrons are transferred to the carpet.
- D** Positive electrons are transferred to the shoe.

10. M/J/23/12/Q27

A metal conductor is connected between the positive and negative terminals of a battery.

Which row gives the direction of movement of the particles that flow in the conductor and their name?

	direction of movement	name of particles
A	from positive terminal to negative terminal	protons
B	from positive terminal to negative terminal	electrons
C	from negative terminal to positive terminal	protons
D	from negative terminal to positive terminal	electrons

11. M/J/23/12/Q28

Three identical cells are connected in parallel to a resistor.

What is the advantage of using three cells in parallel rather than using a single cell?

- A** Each cell produces more energy.
- B** Each cell supplies more charge.
- C** Each cell takes longer to discharge.
- D** The e.m.f. is greater than that of a single cell.

12. M/J/23/12/Q29

A student has three $15\ \Omega$ resistors.

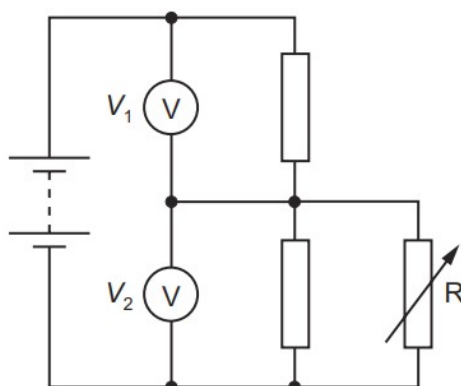
She connects the resistors together in different arrangements.

What is the minimum resistance and what is the maximum resistance of the arrangements that the student can produce?

	minimum resistance / Ω	maximum resistance / Ω
A	5.0	15
B	5.0	45
C	15	30
D	15	45

13. M/J/23/12/Q30

The circuit diagram shows a variable resistor R connected in parallel to the lower section of a potential divider.



The resistance of R increases.

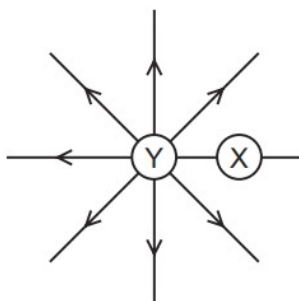
What happens to the two voltmeter readings V_1 and V_2 ?

	V_1	V_2
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

14. M/J/23/11/Q28

Object X is stationary and positively charged. It experiences a force due to the field produced by object Y .

The arrows show the direction of the field produced by Y .



Which statement about the direction of the force on X is correct?

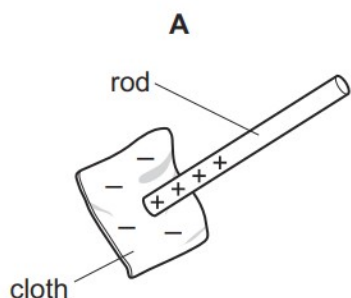
- A** It is towards the right because it is in an electric field.
- B** It is towards the left because it is in an electric field.
- C** It is towards the right because it is in a magnetic field.
- D** It is towards the left because it is in a magnetic field.

15. M/J/23/11/Q27

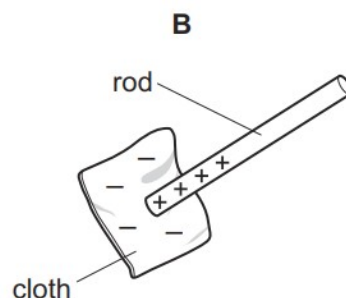
In an electrostatics experiment, a plastic rod is rubbed with a cloth.

The **cloth** becomes negatively charged.

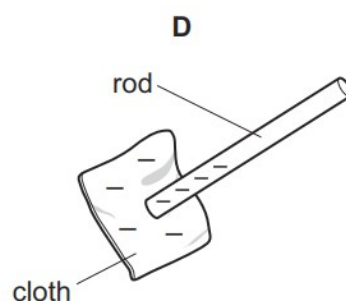
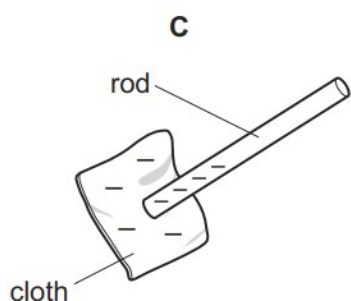
Which diagram shows the charge on the rod and describes the movement of charge?



electrons move from the rod on to the cloth

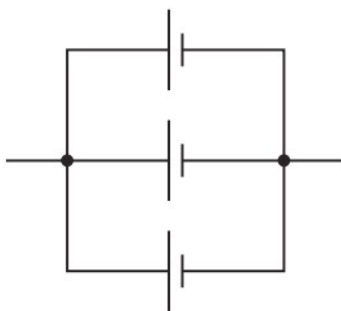


protons move from the cloth on to the rod



16. M/J/23/11/Q29

The diagram shows a battery of three 1.5 V cells.

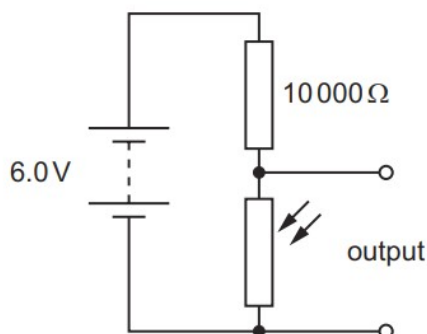


What is an advantage of this arrangement of cells?

- A** The battery can supply a current for a longer time than a single 1.5 V cell.
- B** The battery can supply any e.m.f. between 0 V and 4.5 V.
- C** The battery supplies more energy to each coulomb of charge than a single 1.5 V cell.
- D** The e.m.f. of the battery is 4.5 V.

17. M/J/23/11/Q30

A potential divider consists of a light-dependent resistor (LDR) and a resistor connected to a 6.0 V battery.



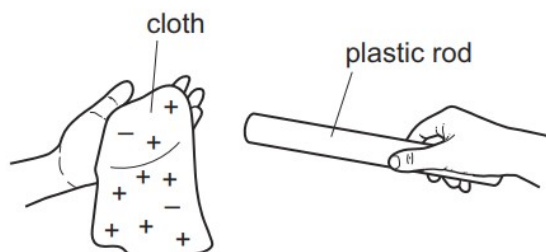
The output is 3.0 V.

What is the resistance of the LDR?

- A** 0
- B** between 0 and 10000Ω
- C** 10000Ω
- D** more than 10000Ω

18. O/N/22/11/Q27

A plastic rod and a cloth are initially uncharged. The rod is then rubbed with the cloth and the cloth becomes positively charged.



What is the sign of the charge on the plastic rod and why does it become charged?

	charge on rod	reason why rod becomes charged
A	negative	electrons move from the cloth to the rod
B	negative	protons move from the rod to the cloth
C	positive	electrons move from the rod to the cloth
D	positive	protons move from the cloth to the rod

19. O/N/22/11/Q28

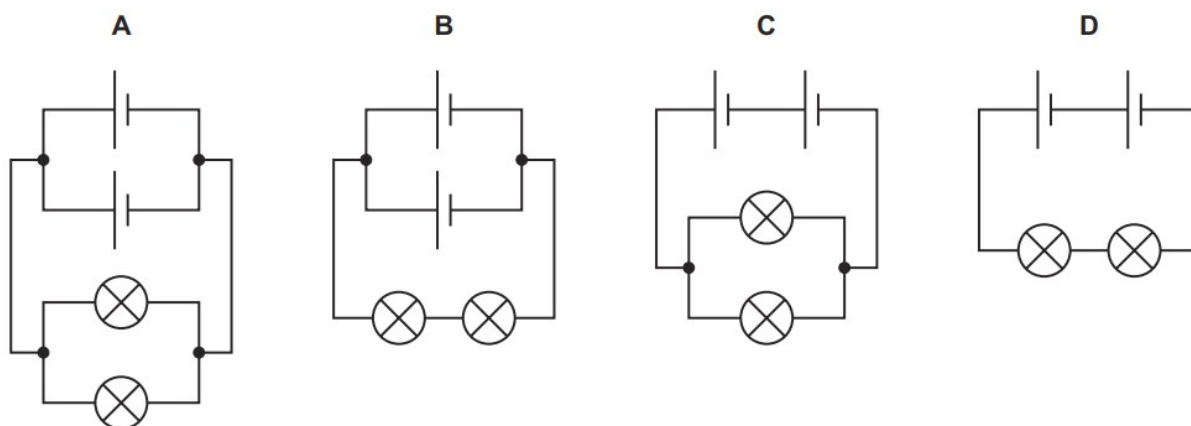
Which expression gives the electromotive force (e.m.f.) of a battery?

- A work done $\times$ time
- B work done $\times$ charge
- C $\frac{\text{work done}}{\text{time}}$
- D $\frac{\text{work done}}{\text{charge}}$

20. O/N/22/11/Q29

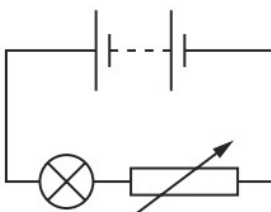
A circuit contains two cells and two lamps. The cells are new and are fully charged.

In which arrangement of the cells and lamps do the cells discharge the most quickly?



21. O/N/22/11/Q30

A lamp is connected in series with a battery and a variable resistor. The resistance of the variable resistor is decreased gradually.



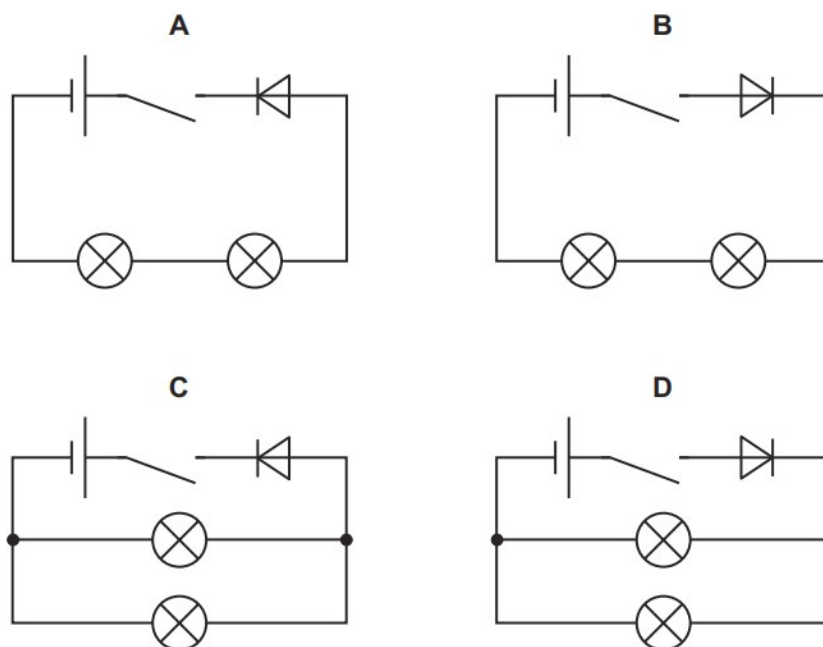
Which statement is correct?

- A The current in the lamp increases.
- B The current in the variable resistor decreases.
- C The e.m.f. of the battery decreases.
- D The total resistance in the circuit increases.

22. O/N/22/11/Q31

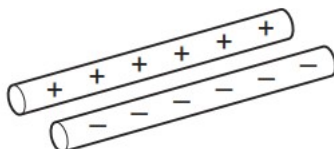
In the circuits shown all the cells are identical and all the lamps are identical. The switches are closed.

In which circuit are both lamps the brightest?



23. O/N/22/12/Q31

Two oppositely charged plastic rods are placed next to each other.



Which row describes and explains how the rods move?

	movement	explanation
A	away from each other	opposite charges attract
B	away from each other	opposite charges repel
C	towards each other	opposite charges attract
D	towards each other	opposite charges repel

24. O/N/22/12/Q32

A meter that measures the rate of flow of charge in a resistor is placed in a circuit.

What is the name of this meter and how is it connected to the resistor?

	name	connection
A	ammeter	in parallel
B	ammeter	in series
C	voltmeter	in parallel
D	voltmeter	in series

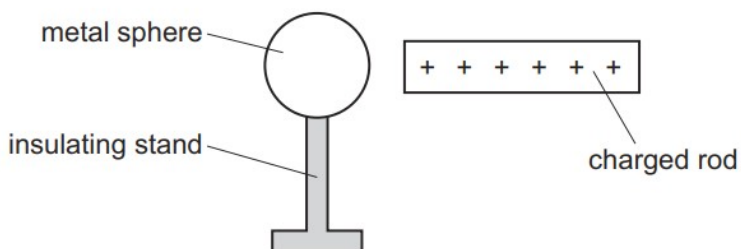
25. O/N/22/12/Q33

Which device is used to convert electrical energy to kinetic energy?

- A** iron
- B** kettle
- C** lamp
- D** motor

26. M/J/22/12/Q30

A positively charged rod is held near an uncharged metal sphere on an insulating stand.

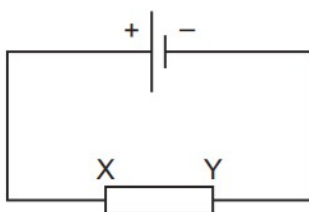


What is the distribution of charge on the sphere?



27. M/J/22/12/Q32

The diagram shows a simple electric circuit.



Which row describes the charge on an electron and the direction of electron flow through the resistor?

	charge on an electron	direction of electron flow
A	negative	X to Y
B	negative	Y to X
C	positive	X to Y
D	positive	Y to X

28. M/J/22/12/Q33

Charge of 0.40 C passes through a resistor in 1.0 s.

In 2.0 s, 20 J of thermal energy is produced in the resistor.

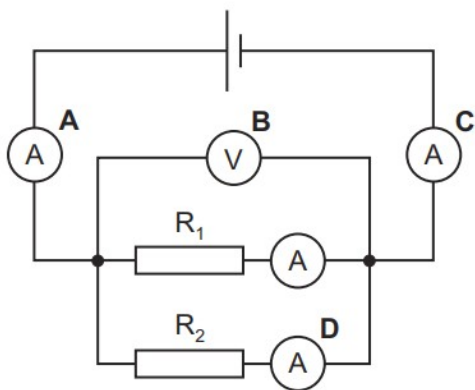
What is the potential difference across the resistor?

- A** 0.020 V **B** 0.040 V **C** 25 V **D** 50 V

29. M/J/22/12/Q34

The diagram shows a simple electric circuit.

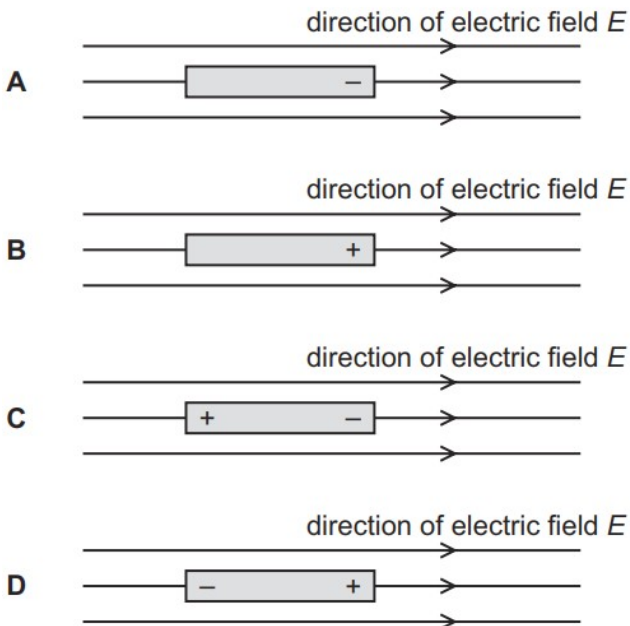
Which meter measures the current in resistor R_2 ?



30. M/J/22/11/Q30

An initially uncharged copper rod is placed in a uniform electric field E . The rod is parallel to the field.

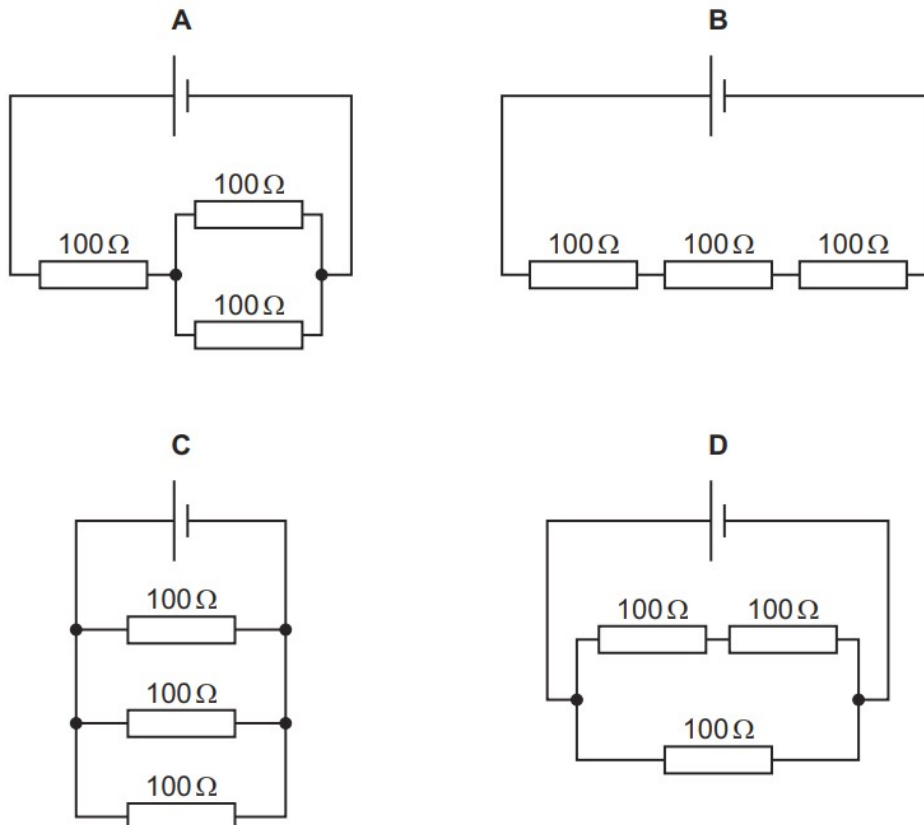
Which diagram shows the charges induced on the rod?



31. M/J/22/11/Q34

The diagrams show four circuits. Each circuit contains three $100\ \Omega$ resistors.

Which circuit has the smallest total resistance?



32. M/J/22/11/Q33

What is the unit of potential difference (p.d.) and which measuring instrument is used to measure p.d.?

	unit	instrument
A	ampere	ammeter
B	ampere	voltmeter
C	volt	ammeter
D	volt	voltmeter

33. O/N/21/11/Q29

A piece of paper torn out of an exercise book is shown.

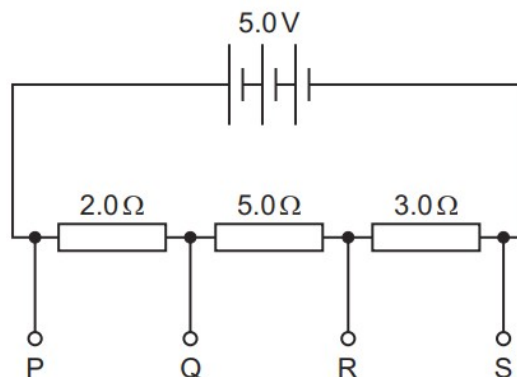
	is the work done (measured in joules)
	on unit charge (measured in coulombs) as it
	moves around the circuit.

What is being defined?

- A** electrical resistance
- B** electric current
- C** electrical power
- D** electromotive force

34. O/N/21/11/Q31

The diagram shows a circuit with a 5.0 V power supply, three resistors and four output terminals P, Q, R and S.

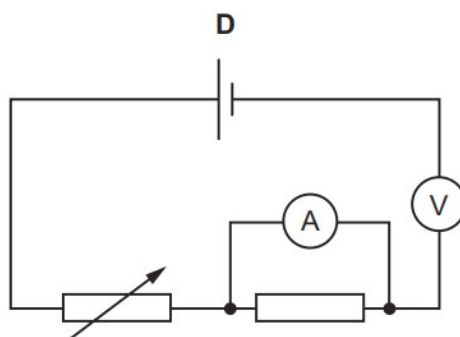
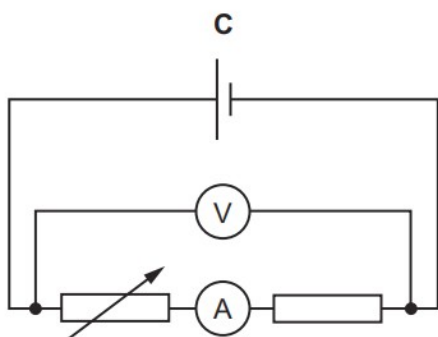
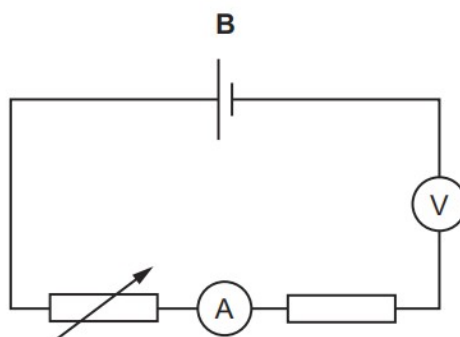
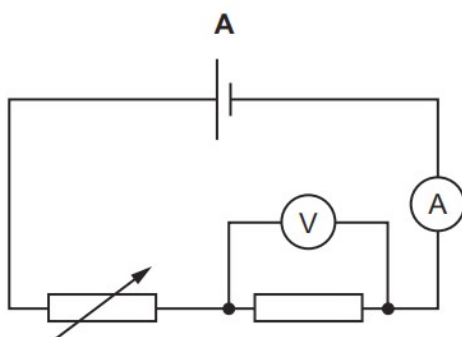


Between which pair of terminals is there a potential difference of 1.5 V?

- A** PQ
- B** PR
- C** QS
- D** RS

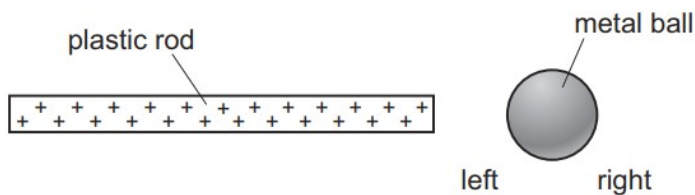
35. O/N/21/11/Q30

Which circuit is connected to measure the current in the fixed resistor and the potential difference (p.d.) across the same resistor?



36. O/N/21/12/Q31

A positively charged plastic rod is held near but not touching an insulated and previously uncharged metal ball. No charge leaves the rod.



What are possible charges on the left-hand side and right-hand side of the ball?

	charge on left-hand side of ball	charge on right-hand side of ball
A	$-2.0 \times 10^{-9} \text{ A}$	$-1.0 \times 10^{-9} \text{ A}$
B	$-2.0 \times 10^{-9} \text{ A}$	$+2.0 \times 10^{-9} \text{ A}$
C	$-2.0 \times 10^{-9} \text{ C}$	$+2.0 \times 10^{-9} \text{ C}$
D	$+2.0 \times 10^{-9} \text{ C}$	$-2.0 \times 10^{-9} \text{ C}$

37. O/N/21/12/Q33

Quantity X and quantity Y have the same unit Z.

X is the energy dissipated by a battery in driving unit charge round the complete circuit.

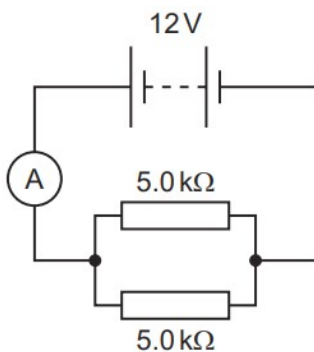
Y is the work done in a component in driving unit charge through the component.

What are X, Y and Z?

	X	Y	Z
A	electromotive force	potential difference	coulomb
B	electromotive force	potential difference	volt
C	potential difference	electromotive force	coulomb
D	potential difference	electromotive force	volt

38. M/J/21/12/Q32

The diagram shows a circuit.



The ammeter has four different range settings.

Which range setting gives a reading near the mid-point of the range?

- A** 0–1 mA **B** 0–10 mA **C** 0–1.0 A **D** 0–10 A

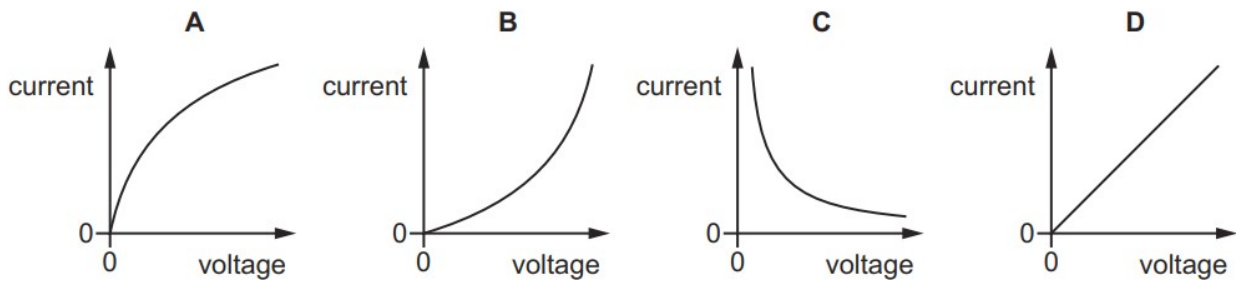
39. M/J/21/12/Q33

In which example is the current 1.0 A?

- A** a charge of 0.10 C passes through a resistor in 100 ms
B a 10 V lamp with input power 0.10 W
C a resistor of resistance 0.50Ω with a potential difference (p.d.) of 2.0 V across it
D two resistors with a current of 0.50 A connected in series

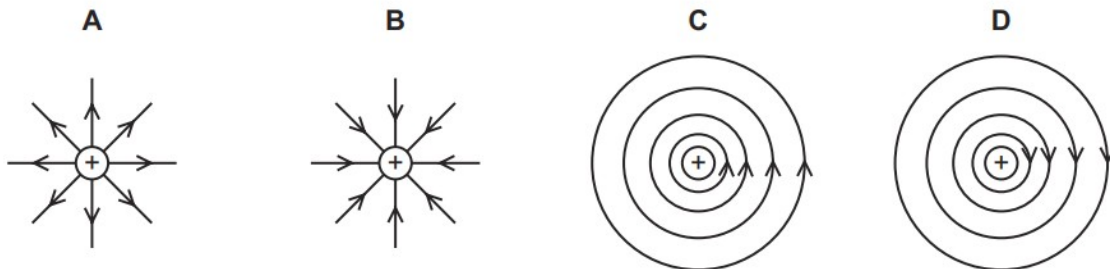
40. M/J/21/12/Q34

Which graph shows the relationship between current and voltage for a filament lamp?



41. M/J/21/11/Q27

Which diagram shows the pattern and direction of the electric field lines near a positive charge?



42. O/N/20/11/Q34

A small negative charge is placed at a point where an electric field is vertically upwards.

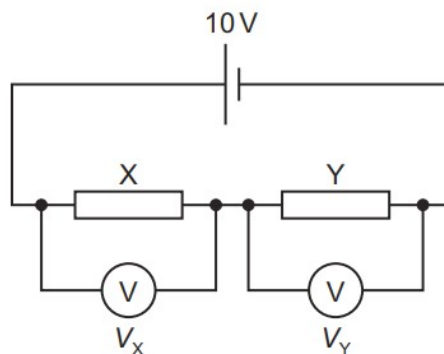
There is a force on the charge due to the field.

In which direction does it act?

- A** vertically upwards
- B** vertically downwards
- C** horizontally to the right
- D** horizontally to the left

43. O/N/20/11/Q35

In the circuit shown, the resistance of resistor Y is four times greater than the resistance of resistor X.



What is the difference $V_Y - V_X$ of the voltages shown on the voltmeters?

- A** 2.0V
- B** 5.0V
- C** 6.0V
- D** 8.0V

44. O/N/20/11/Q36

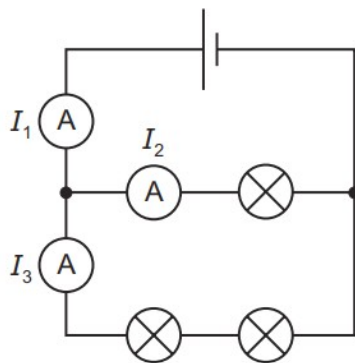
Two identical resistors connected in parallel have a total resistance of $4.0\ \Omega$.

What is the total resistance when the same two resistors are connected in series?

- A** $1.0\ \Omega$ **B** $4.0\ \Omega$ **C** $8.0\ \Omega$ **D** $16\ \Omega$

45. O/N/20/11/Q37

Three identical lamps and three ammeters are connected as shown.



The readings on the ammeters are I_1 , I_2 and I_3 .

How are the readings related?

- A** $I_1 = I_2 = I_3$
B $I_1 > I_2$ and $I_2 = I_3$
C $I_1 > I_3 > I_2$
D $I_1 > I_2 > I_3$

46. O/N/20/12/Q29

What is stored in a battery and what is its unit?

	quantity	unit
A	current	A
B	current	As
C	energy	J
D	energy	J/s

47. O/N/20/12/Q30

A car battery delivers $150\ \text{kC}$ to a circuit before it needs recharging.

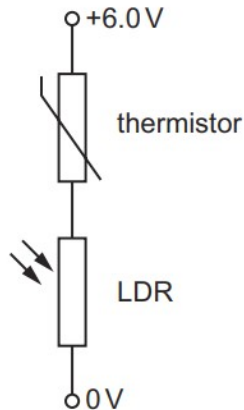
Two headlamps connected in parallel to the battery are switched on. There is a current of $4.0\ \text{A}$ in each lamp.

How much time passes before the battery needs recharging?

- A** 2.6 hours **B** 5.2 hours **C** 7.8 hours **D** 10.4 hours

48. O/N/20/11/Q38

A thermistor and a light-dependent resistor (LDR) are connected in series. A potential difference (p.d.) of 6.0 V is applied across them as shown.



The thermistor has a resistance of $6000\ \Omega$ in a cold room and $1000\ \Omega$ in a warm room. The LDR has a resistance of $2000\ \Omega$ in dim light and $500\ \Omega$ in bright light.

When is the p.d. across the LDR equal to 2.0 V?

- A** in a cold room with bright light
- B** in a cold room with dim light
- C** in a warm room with bright light
- D** in a warm room with dim light

49. M/J/20/12/Q36

A stationary negative charge in an electric field experiences an electric force in the direction shown.



What is the direction of the electric field?

- A** to the left
- B** to the right
- C** down the page
- D** up the page

50. M/J/20/12/Q37

A metal wire of length l and cross-sectional area A has resistance R .

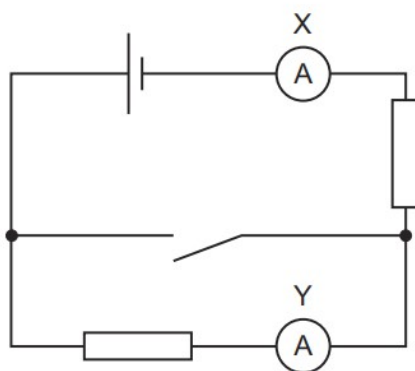
A second wire is made from the same metal. It has a length $2l$ and a cross-sectional area $4A$.

What is the resistance of the second wire?

- A** $8R$
- B** $2R$
- C** $\frac{R}{2}$
- D** $\frac{R}{8}$

51. M/J/20/12/Q38

The diagram shows a circuit.

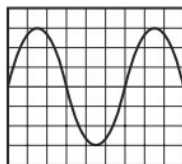


What effect does closing the switch have on the readings of ammeters X and Y?

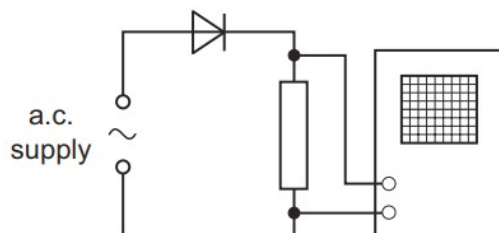
	reading on ammeter X	reading on ammeter Y
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

52. O/N/19/11/Q33

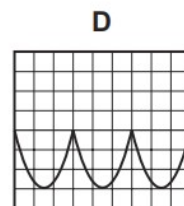
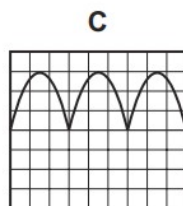
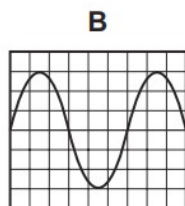
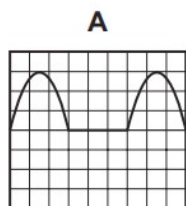
An a.c. supply connected to an oscilloscope produces the trace shown.



A diode is now connected in series to the circuit as shown.



What is the new trace on the screen?



53. O/N/19/11/Q34

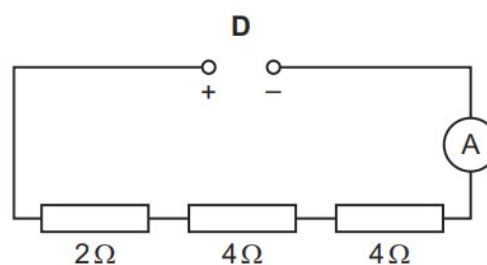
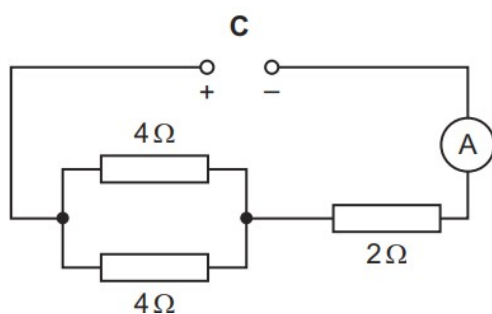
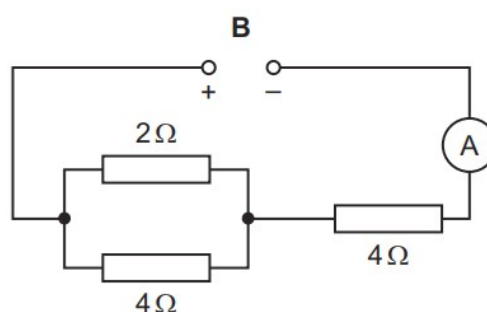
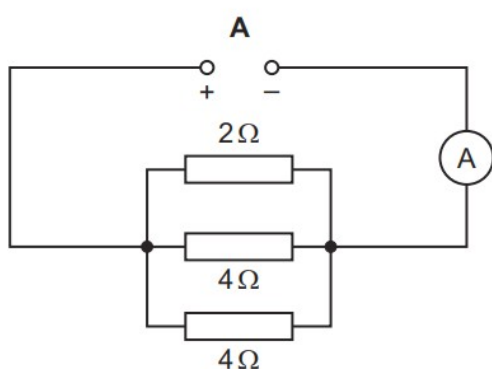
Resistors of different resistances are connected to a power supply in either a parallel circuit or a series circuit. Which statement is correct?

- A The current is the same in all resistors connected in parallel.
- B The current is the same in all resistors connected in series.
- C The voltage across each resistor is different for all resistors connected in parallel.
- D The voltage across each resistor is the same for all resistors connected in series.

54. O/N/19/11/Q35

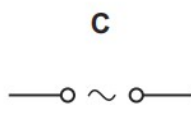
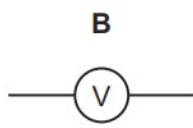
An ammeter is connected to three resistors and a power supply.

Which arrangement of resistors gives the greatest ammeter reading?



55. O/N/19/12/Q32

Which symbol does **not** represent a power source?



56. O/N/19/12/Q35

A hotplate on an electric cooker contains two identical resistors.

The switch has three positions.

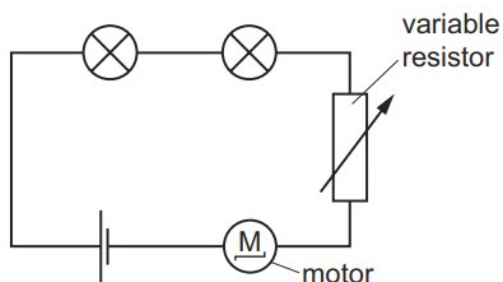
- position 1 The two resistors are connected in series to the mains supply.
position 2 The two resistors are connected in parallel to the mains supply.
position 3 Just one resistor is connected to the mains supply.

Which positions correspond to the low, the medium and the high power settings?

	low power	medium power	high power
A	1	2	3
B	1	3	2
C	3	1	2
D	3	2	1

57. M/J/19/12/Q33

The diagram shows a simple circuit.

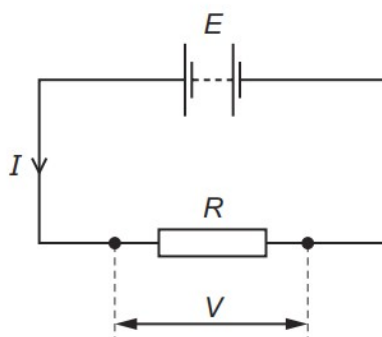


What happens when the resistance of the variable resistor is increased?

- A** The lamps are dimmer and the motor rotates more quickly.
B The lamps are dimmer and the motor rotates more slowly.
C The lamps have the same brightness and the motor rotates more quickly.
D The lamps have the same brightness and the motor rotates more slowly.

58. M/J/19/12/Q34

The diagram shows a simple electrical circuit.

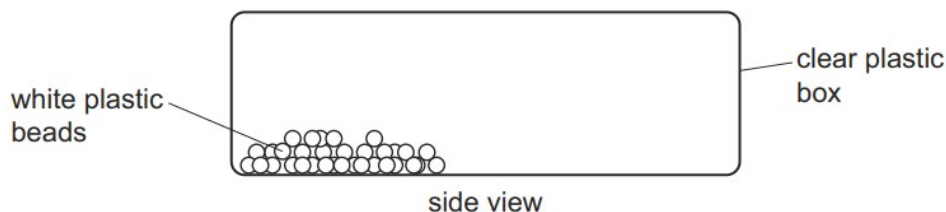


Which statement **cannot** be correct?

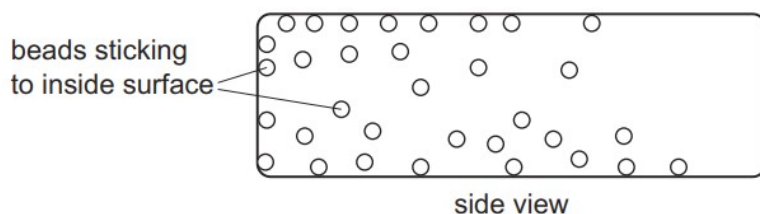
- A The current I is 2 A.
- B The e.m.f. E is 12 J.
- C The p.d. V (across the resistor) is 10 V.
- D The resistance R is 5Ω .

59. M/J/19/11/Q31

The diagram shows some white plastic beads in a clear plastic box.



The box is shaken and the beads rub against the box. The beads stick to the inside surface of the box.



Which row is a possible explanation for this?

	box	beads
A	gains electrons	no change of electrons
B	loses electrons	gain electrons
C	no change of electrons	lose electrons
D	no change of electrons	gain electrons

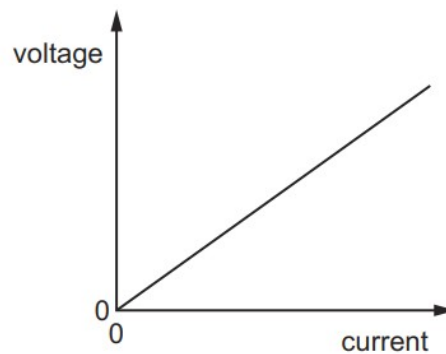
60. M/J/19/11/Q32

Which unit is the same as a volt?

- A ampere / ohm
- B ampere / watt
- C ohm / ampere
- D watt / ampere

61. M/J/19/11/Q33

The voltage / current graph for a metal wire is shown.



What does the gradient of this graph represent?

- A the charge passing through the wire
- B the e.m.f. of the battery connected to the wire
- C the energy produced in the wire
- D the resistance of the wire

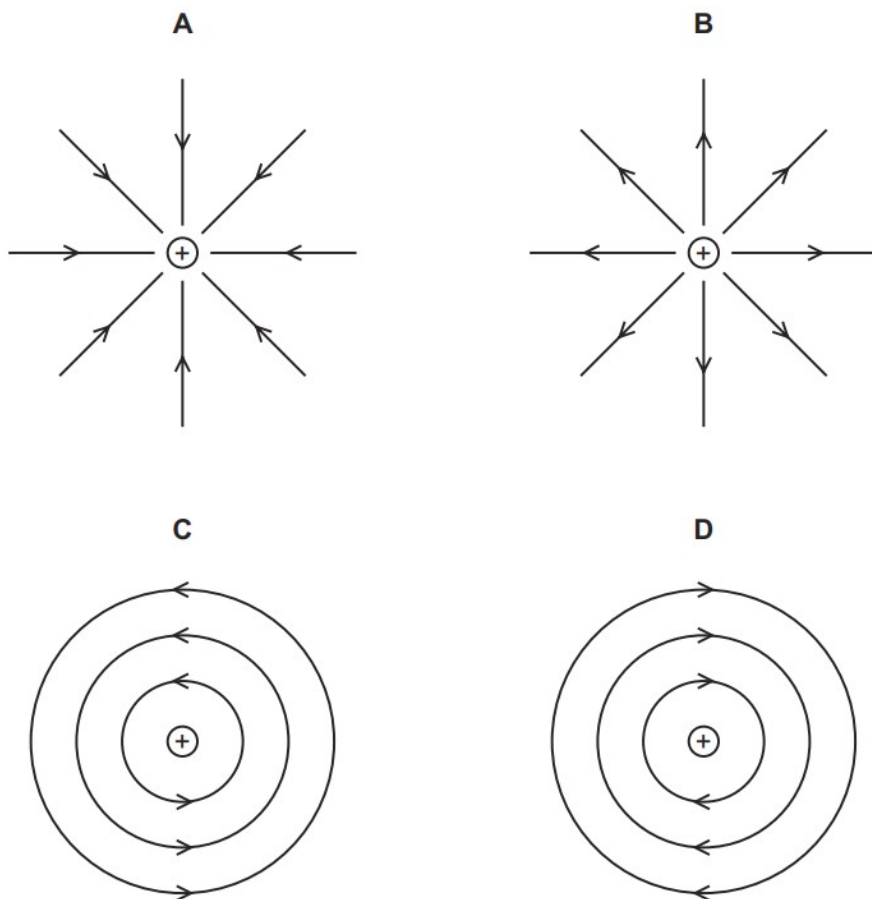
62. M/J/19/11/Q34

What is the symbol for a device that measures current?



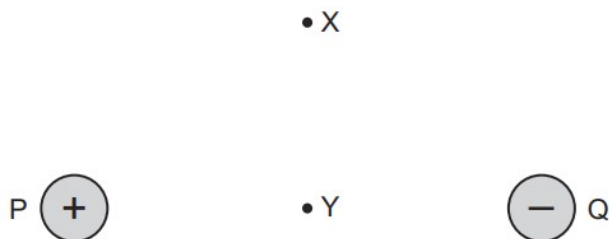
63. O/N/18/11/Q24

Which diagram shows the electric field around an isolated point positive charge?



64. O/N/18/11/Q25

An electric field is formed by two isolated, equal and opposite charges P and Q.



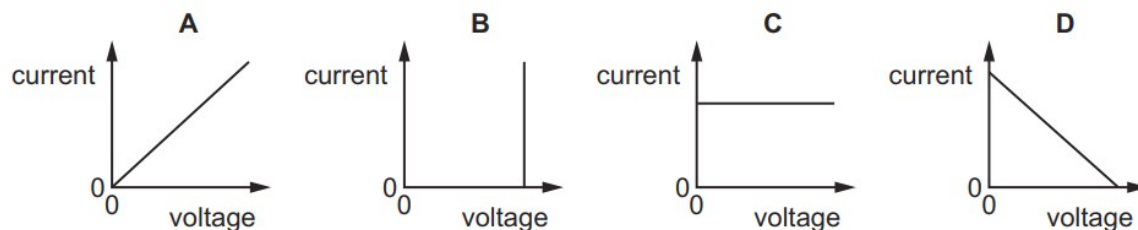
How does the field at X compare with the field at Y?

- A** It is stronger at X than at Y and is in the opposite direction.
- B** It is stronger at X than at Y and is in the same direction.
- C** It is weaker at X than at Y and is in the opposite direction.
- D** It is weaker at X than at Y and is in the same direction.

65. O/N/18/11/Q27

The voltage across a resistor of fixed resistance changes.

Which graph shows how the current in the resistor changes with the voltage?



66. O/N/18/11/Q28

A wire of length 0.50 m and cross-sectional area $1.0 \times 10^{-6} \text{ m}^2$ has a resistance of 0.75Ω .

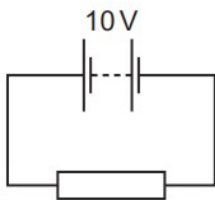
Another wire of the same material has a length of 2.0 m and a cross-sectional area of $0.50 \times 10^{-6} \text{ m}^2$.

What is the resistance of the longer wire?

- A** 0.094Ω **B** 0.38Ω **C** 1.5Ω **D** 6.0Ω

67. O/N/18/11/Q29

A battery of electromotive force (e.m.f.) 10 V is connected to a resistor.



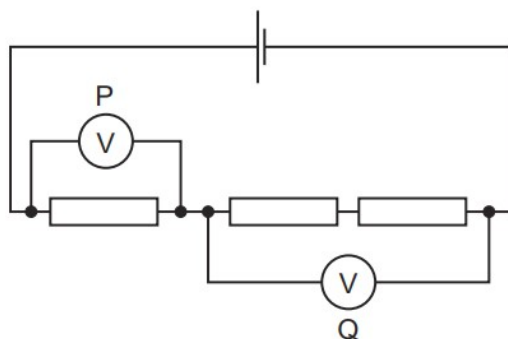
A charge of 2.0 coulombs passes through the resistor.

How much work is done as the charge passes through the resistor?

- A** 0.20 J **B** 2.0 J **C** 5.0 J **D** 20 J

68. O/N/18/11/Q30

Three identical resistors are joined in series to a cell.



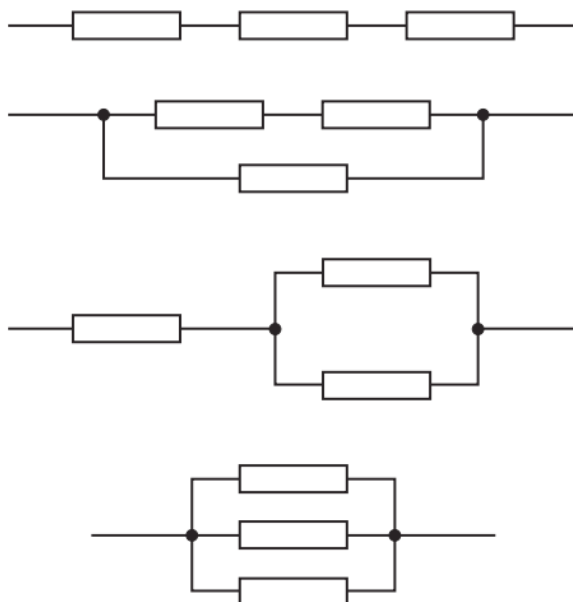
Voltmeter Q reads 8.0 V.

What is the reading on voltmeter P and what is the e.m.f. of the cell?

	reading on P/V	e.m.f. of cell/V
A	4.0	8.0
B	4.0	12
C	8.0	8.0
D	8.0	12

69. O/N/18/12/Q30

The diagram shows three 6.0Ω resistors connected to give various total resistances.



What is the difference between the smallest total resistance and the largest total resistance?

- A** 7.0Ω **B** 9.0Ω **C** 12Ω **D** 16Ω

70. O/N/18/12/Q29

Students are asked to describe an experiment to measure the resistance of a metallic conductor.

Which description is correct?

- A** Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{I}{V}$.
- B** Connect an ammeter in parallel and a voltmeter in series with the conductor then use $R = \frac{V}{I}$.
- C** Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{I}{V}$.
- D** Connect a voltmeter in parallel and an ammeter in series with the conductor then use $R = \frac{V}{I}$.

71. O/N/18/12/Q32

A defibrillator is a device that is used to give an electric shock to a patient's heart.

It supplies an electric shock with energy 240 J at an average voltage of 2000 V for 10 ms.

What is the average current it supplies?

- A** 0.012 A **B** 1.2 A **C** 12 A **D** 120 A

72. M/J/18/12/Q35

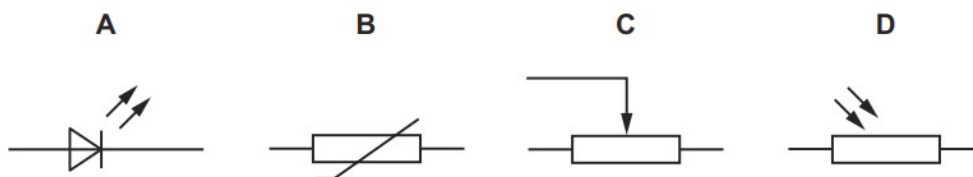
V is a potential difference, I is a current, R is a resistance, and t is a time.

Which expression has units of energy?

- A** It **B** I^2R **C** VIt **D** $\frac{V^2}{R}$

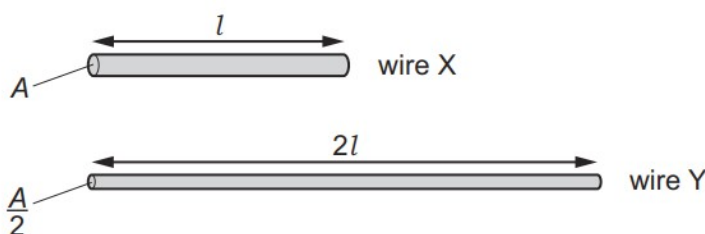
73. M/J/18/12/Q37

Which symbol is used for a light-dependent resistor?



74. M/J/18/11/Q29

A copper wire X has resistance R . Another copper wire Y has twice the length and half the cross-sectional area of X.

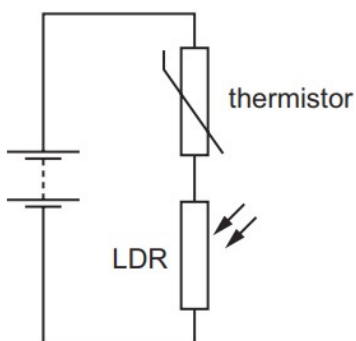


What is the resistance of Y?

- A** $\frac{R}{2}$ **B** R **C** $2R$ **D** $4R$

75. M/J18/11/Q30

In the circuit shown, the temperature of the room and the amount of light affect the current.



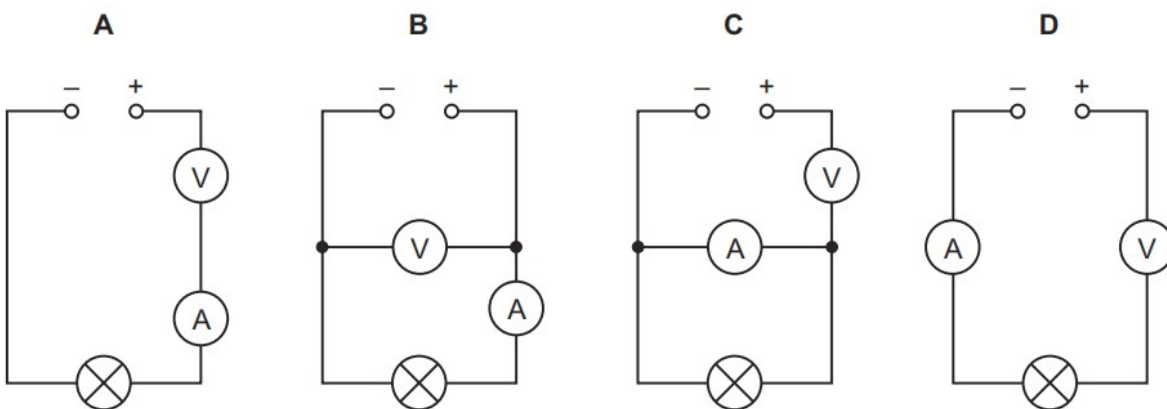
Under which conditions is the current in the circuit the largest?

	temperature	amount of light
A	high	in bright light
B	high	in the dark
C	low	in bright light
D	low	in the dark

76. O/N/17/11/Q30

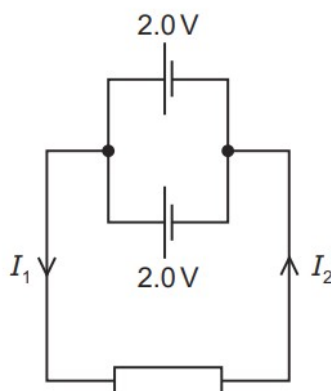
In a circuit, a voltmeter is used to measure the potential difference across a lamp. An ammeter is used to measure the current in the lamp.

Which diagram shows the circuit?



77. O/N/17/11/Q31

The battery in a circuit contains two identical cells connected in parallel.



Which row shows the relationship between currents I_1 and I_2 and gives the electromotive force (e.m.f.) of the battery?

	relationship	e.m.f. / V
A	$I_1 > I_2$	2.0
B	$I_1 > I_2$	4.0
C	$I_1 = I_2$	2.0
D	$I_1 = I_2$	4.0

78. O/N/17/12/Q30

A piece of wire has a resistance of $16\ \Omega$.

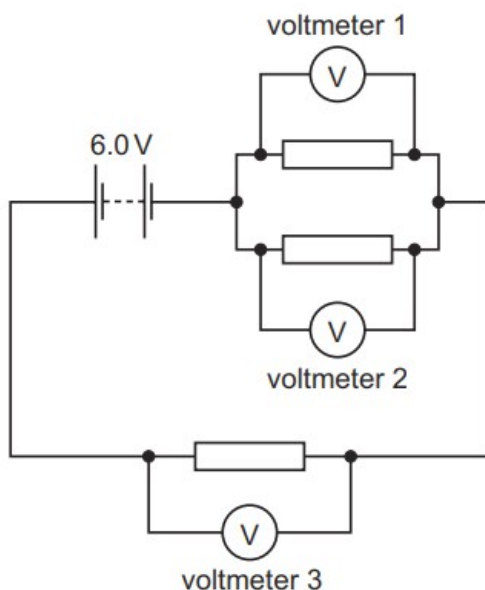
The wire is 20 cm long and has a cross-sectional area of $2.0\ \text{mm}^2$.

Which wire of the same material has a resistance of $8.0\ \Omega$?

	length / cm	cross-sectional area / mm^2
A	10	1.0
B	10	4.0
C	20	1.0
D	20	4.0

79. O/N/17/12/Q29

A 6.0 V battery, three resistors and three voltmeters are connected in the circuit shown.



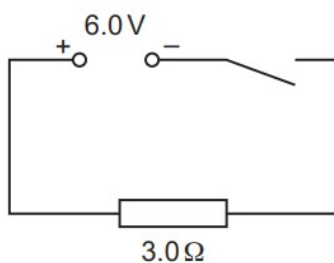
The reading on voltmeter 1 is 2.0 V.

What is the reading on voltmeter 2 and what is the reading on voltmeter 3?

	reading on voltmeter 2 /V	reading on voltmeter 3 /V
A	2.0	4.0
B	2.0	8.0
C	4.0	4.0
D	4.0	8.0

80. M/J/17/12/Q33

The circuit shown is switched on for 1.0 minute.



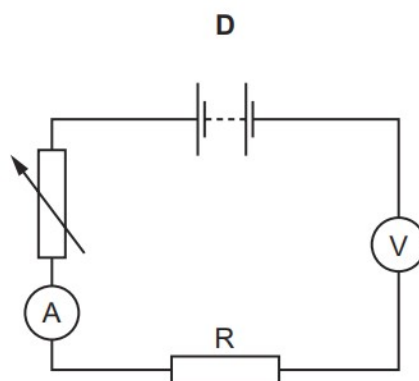
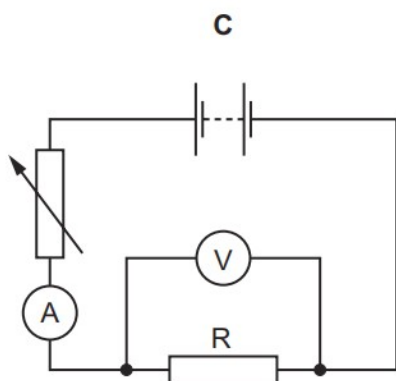
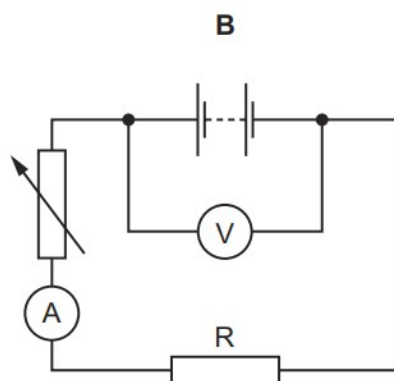
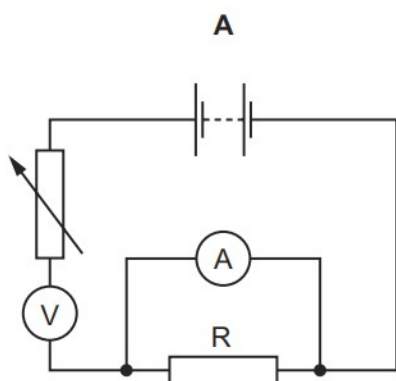
How much charge passes through the 3.0 Ω resistor?

- A** 2.0 C **B** 12 C **C** 120 C **D** 720 C

81. M/J/17/12/Q34

The readings of an ammeter and a voltmeter are used to calculate the resistance of resistor R.

Which circuit shows the ammeter and voltmeter in the correct positions?



82. M/J/17/11/Q36

During a thunderstorm, a lightning flash occurs when charge flows between a cloud and the Earth. On average, 20 C of charge flows during each flash and there are three flashes every minute.

What is the average current between the cloud and the Earth during the thunderstorm?

- A** 1.0 A **B** 36 A **C** 60 A **D** 1800 A

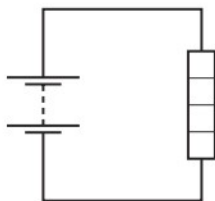
83. M/J/17/11/Q37

Why are batteries sometimes made from several identical cells arranged in parallel?

- A** The battery has a greater e.m.f. than a single cell on its own.
- B** The battery has a lower e.m.f. than a single cell on its own.
- C** The battery lasts longer than a single cell on its own.
- D** The battery supplies a smaller current than a single cell on its own.

84. O/N/16/12/Q33

An ammeter and a voltmeter are connected into the circuit shown when determining the power produced by a heater.



How are these meters connected, in relation to the heater?

	ammeter	voltmeter
A	in parallel	in parallel
B	in parallel	in series
C	in series	in parallel
D	in series	in series

85. M/J/16/12/Q25

A charge of 45 C flows through an electric appliance in 3.0 minutes.

What is the average current in the appliance?

- A** 0.25 A **B** 4.0 A **C** 15 A **D** 135 A

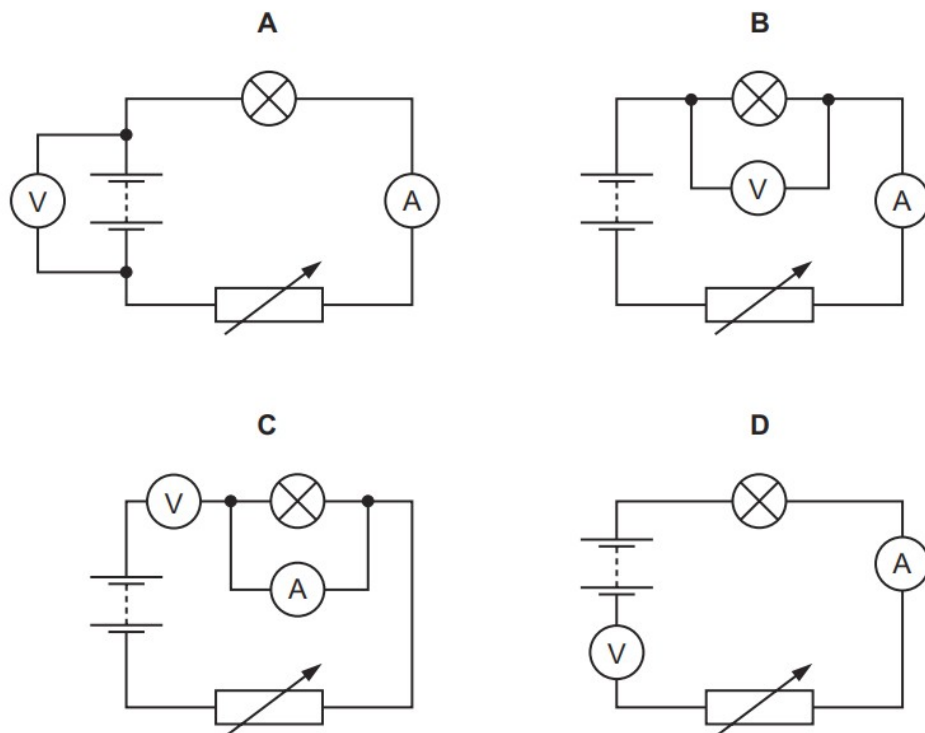
86. M/J/16/12/Q26

How can *one volt* also be expressed?

- A** one coulomb per ampere
B one coulomb per joule
C one joule per ampere
D one joule per coulomb

87. M/J/16/12/Q28

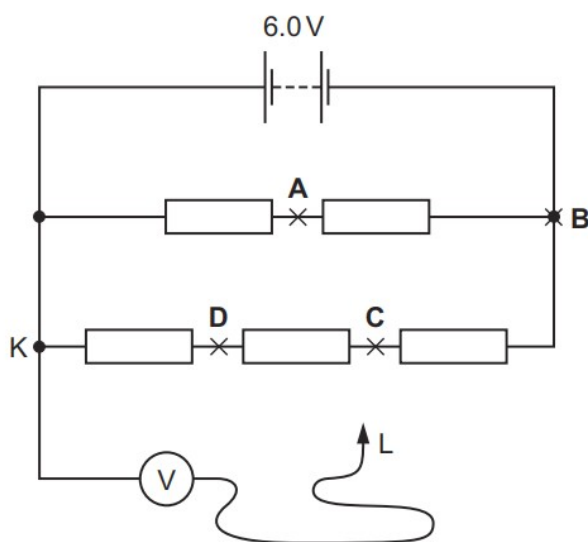
Which diagram shows the arrangement of the ammeter and voltmeter to obtain readings to find the power of a lamp?



88. M/J/16/12/Q29

A 6.0V battery is connected to a network containing five identical resistors. A voltmeter has one lead connected to point K as shown.

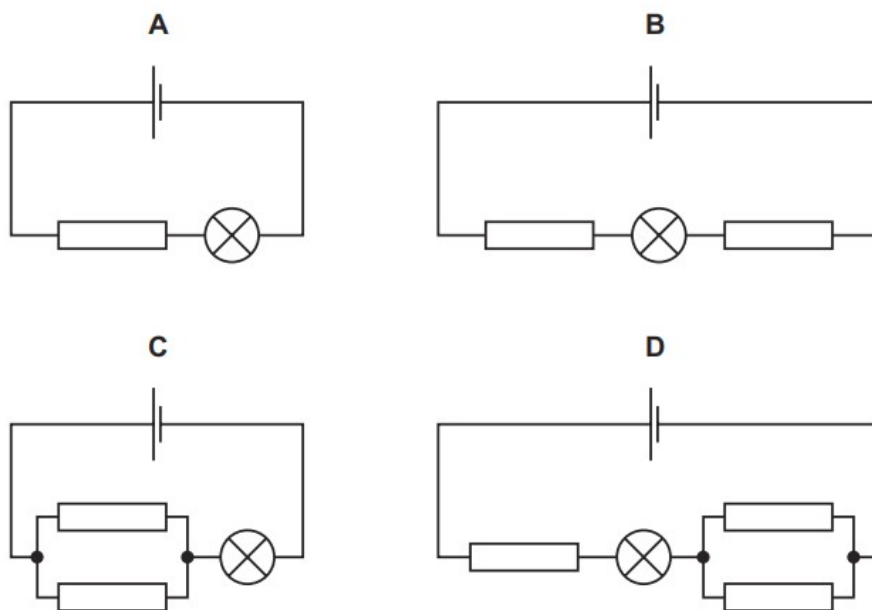
At which point should lead L be connected so that the voltmeter reads 3.0V?



89. M/J/16/11/Q31

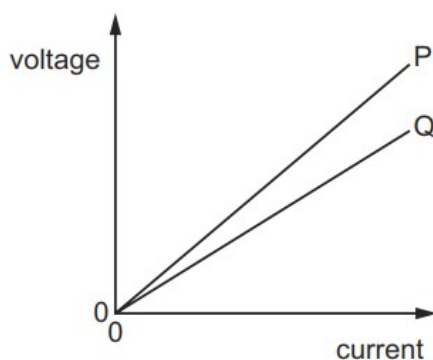
The cells, lamps and resistors in the circuits are identical.

In which circuit is the lamp the brightest?



90. M/J/16/11/Q32

The graph is the voltage-current graph for two resistance wires P and Q.



The wires are made from the same material and have equal lengths.

The resistances of the wires and their cross-sectional areas are different.

Which wire has the greater resistance and which wire has the larger cross-sectional area?

	greater resistance	larger cross-sectional area
A	P	P
B	P	Q
C	Q	P
D	Q	Q

91. O/N/15/11/Q24

A polythene rod becomes charged when rubbed with a dry woollen cloth.

The charge caused is a result of

- A** friction producing a movement of electrons.
- B** friction producing a movement of protons.
- C** magnetism producing a movement of electrons.
- D** magnetism producing a movement of protons.

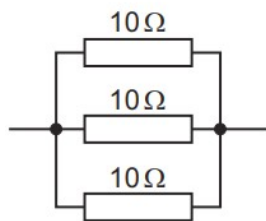
92. O/N/15/11/Q27

Which physical quantity is produced by a calculation where a charge is multiplied by a potential difference (p.d.)?

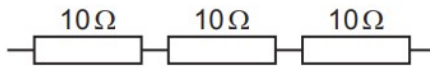
- A** current
- B** electromotive force (e.m.f.)
- C** energy
- D** power

93. O/N/15/11/Q28

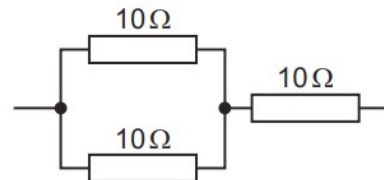
Three 10Ω resistors are connected together to form networks X, Y and Z.



network X



network Y



network Z

What is the order of resistance of the networks going from the smallest total resistance to the largest total resistance?

- A** $X \rightarrow Y \rightarrow Z$ **B** $X \rightarrow Z \rightarrow Y$ **C** $Z \rightarrow X \rightarrow Y$ **D** $Z \rightarrow Y \rightarrow X$

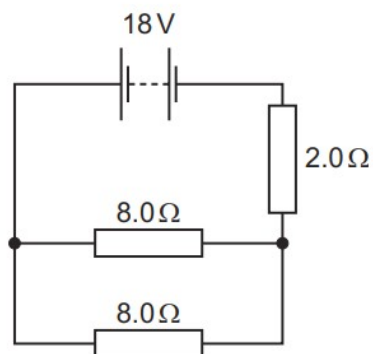
94. O/N/15/12/Q29

Which row correctly names an electrical conductor and an insulator?

	electrical conductor	insulator
A	aluminium	iron
B	iron	rubber
C	plastic	aluminium
D	rubber	plastic

95. O/N/15/12/Q33

A power supply of 18 V is connected to three resistors, as shown.



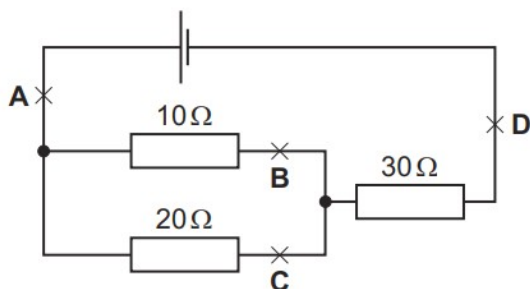
What is the potential difference across the 2.0Ω resistor?

- A** 2.0 V **B** 3.6 V **C** 6.0 V **D** 12 V

96. M/J/15/12/Q33

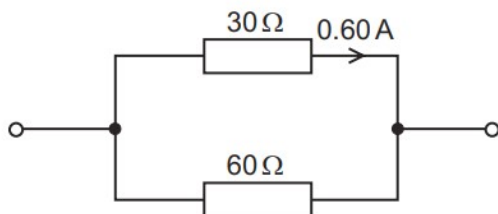
The diagram shows a circuit.

Where must an ammeter be connected to measure the smallest current?



97. M/J/15/12/Q34

Two resistors of resistances 30Ω and 60Ω are arranged in parallel. The current in the 30Ω resistor is 0.60 A.

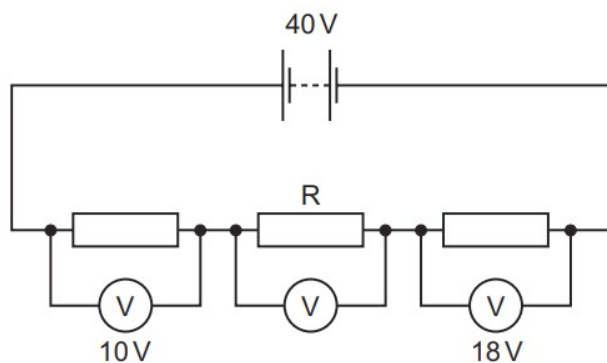


What is the potential difference across the 60Ω resistor?

- A** 9.0 V **B** 18 V **C** 36 V **D** 54 V

98. M/J/15/12/Q35

The circuit shows three resistors in series connected to a battery. Each resistor has a voltmeter across it and two of the voltages are shown.

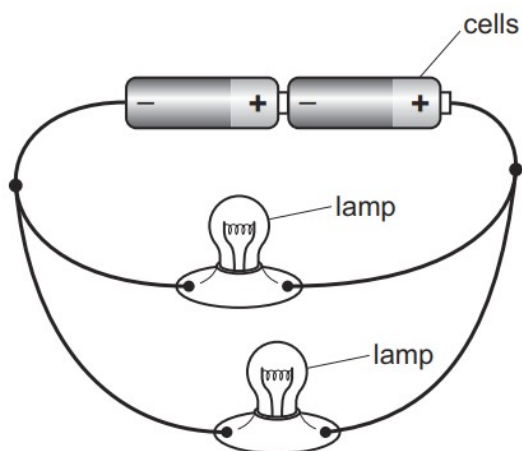


What is the potential difference (p.d.) across the resistor R?

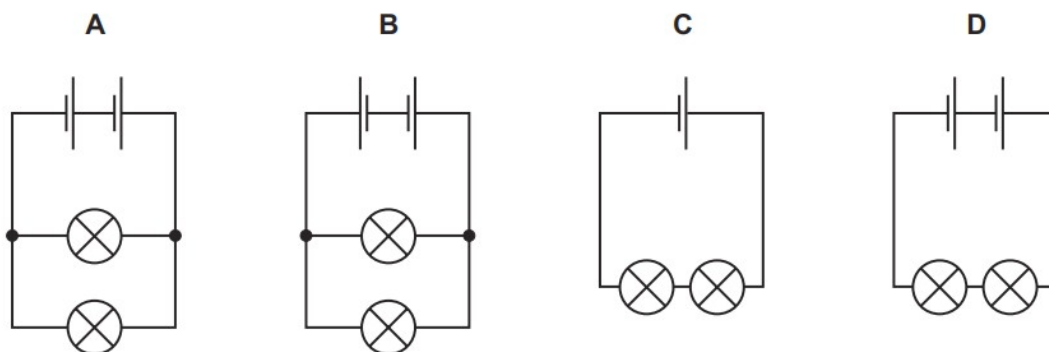
- A** 12V **B** 22V **C** 30V **D** 68V

99. M/J/15/11/Q28

The diagram shows a circuit.



Which circuit diagram shows this circuit?



100. M/J/15/11/Q29

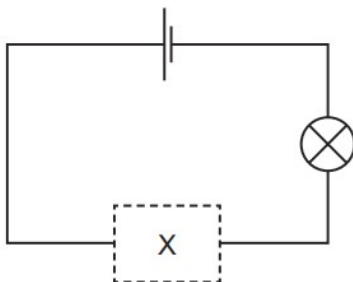
A charge of 7.5 C flows through a resistor in 5.0 s . A student has ammeters with different ranges that he can use to measure the current in the resistor.

Which ammeter range is the most appropriate?

- A** 0-1 A **B** 0-2 A **C** 0-5 A **D** 0-40 A

101. O/N/14/11/Q28

With component X placed in the circuit below, the brightness of the lamp can be gradually reduced until it goes out.



What is component X?

- A** a reed relay
B a switch
C a transformer
D a variable resistor

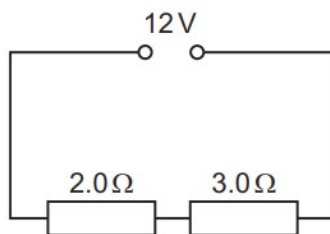
102. O/N/14/12/Q24

Which list contains only electrical insulators?

- A** glass, plastic, rubber
B glass, plastic, steel
C glass, rubber, steel
D plastic, rubber, steel

103. O/N/14/12/Q26

The diagram shows a circuit that has two resistors in series with a 12 V supply.

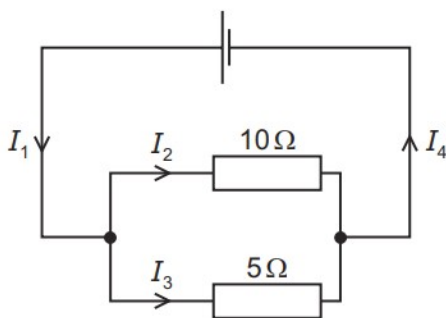


What is the current in the circuit?

- A** 2.4 A **B** 10 A **C** 14 A **D** 60 A

104. O/N/14/12/Q25

The currents in different parts of the circuit are I_1 , I_2 , I_3 and I_4 .



Which statement is correct?

- A** $I_1 = I_4$ and I_2 is greater than I_3 .
- B** $I_1 = I_4$ and I_3 is greater than I_2 .
- C** I_2 is greater than I_1 and less than I_3 .
- D** I_2 is greater than I_1 and greater than I_3 .

105. M/J/14/11/Q30

A piece of wire has a resistance of 16Ω .

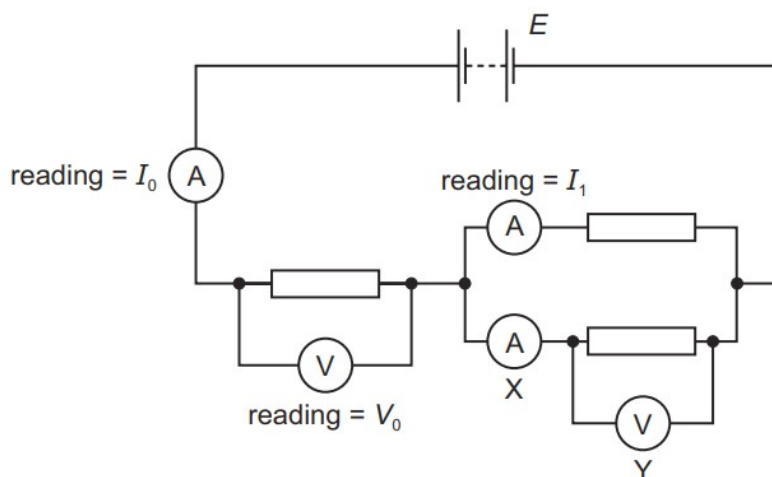
Another wire made from the same metal has four times the length and twice the cross-sectional area.

What is the resistance of the wire?

- A** 8Ω
- B** 32Ω
- C** 96Ω
- D** 128Ω

106. M/J/14/11/Q31

A battery of e.m.f. E is connected in a circuit containing three resistors.



The readings of two ammeters and a voltmeter are shown.

Which readings are shown on ammeter X and on voltmeter Y?

	X	Y
A	$I_0 - I_1$	$E - V_0$
B	$I_0 - I_1$	$E + V_0$
C	$I_0 + I_1$	$E + V_0$
D	$I_0 + I_1$	$E - V_0$

107. O/N/13/11/Q27

Which row shows a conductor and an insulator?

	conductor	insulator
A	aluminium	copper
B	aluminium	glass
C	plastic	copper
D	plastic	glass

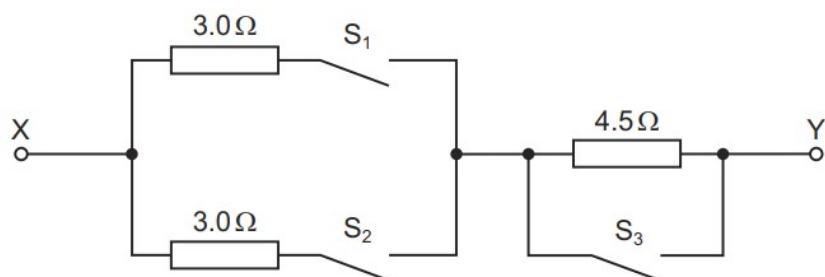
108. O/N/13/11/Q28

Which of the following is equivalent to one coulomb?

- A** one ampere second
- B** one ampere per volt
- C** one volt ampere
- D** one volt per ampere

109. O/N/13/11/Q29

The diagram shows a circuit in which all the switches are open.

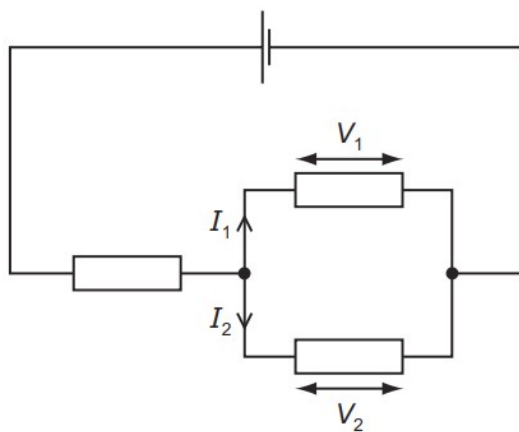


Which switch positions give a resistance of 6.0Ω between X and Y?

	S_1	S_2	S_3
A	closed	closed	closed
B	closed	closed	open
C	closed	open	closed
D	closed	open	open

110. O/N/13/11/Q30

The circuit shows three resistors connected to a cell.



The resistors have different values of resistance.

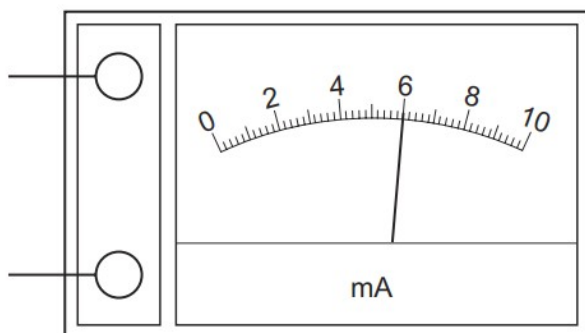
V_1 and V_2 are potential differences and I_1 and I_2 are currents as shown.

Which pair of relationships is correct?

- A** $I_1 = I_2$ and $V_1 = V_2$
- B** $I_1 = I_2$ and $V_1 \neq V_2$
- C** $I_1 \neq I_2$ and $V_1 = V_2$
- D** $I_1 \neq I_2$ and $V_1 \neq V_2$

111. O/N/13/12/Q28

An ammeter is used to measure the current in a $300\ \Omega$ resistor. The ammeter is shown below.

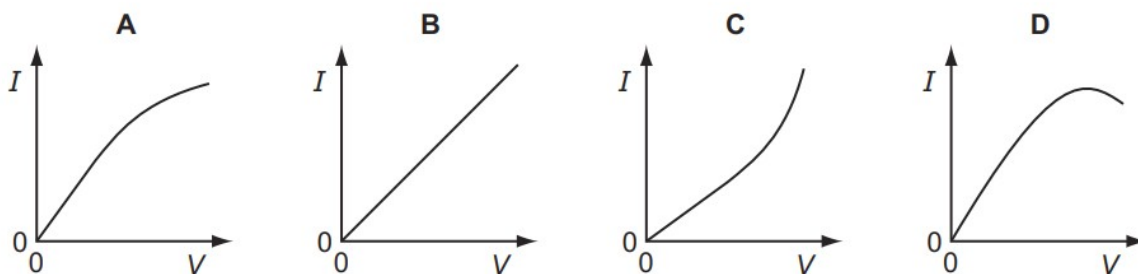


What is the potential difference across the resistor?

- A** 0.050 V **B** 1.8 V **C** 50 V **D** 1800 V

112. O/N/13/12/Q30

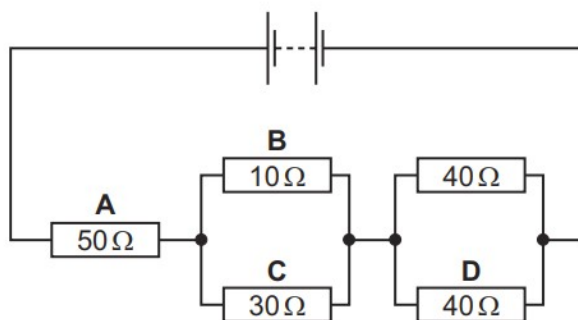
Which is the current/voltage (I/V) graph of a filament lamp?



113. O/N/13/12/Q31

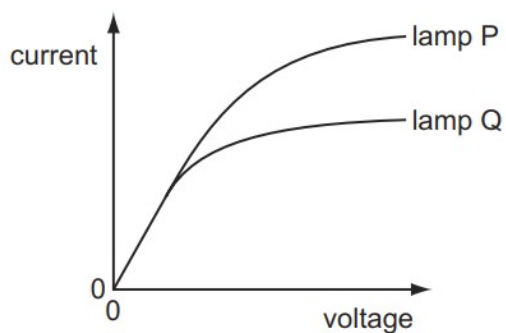
The diagram shows a circuit containing five resistors connected to a battery.

In which resistor is the current the smallest?



114. M/J/13/12/Q33

The diagram shows the current-voltage graph of two filament lamps.

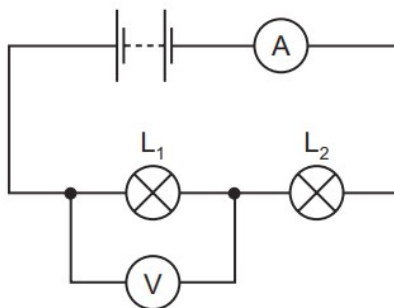


Which statement is correct?

- A** Lamp P has a lower resistance than lamp Q at all currents.
- B** Lamp P has the same resistance as lamp Q at low currents.
- C** Lamp P has the same resistance as lamp Q at high currents.
- D** Lamp P has a higher resistance than lamp Q at all currents.

115. M/J/13/12/Q34

A circuit containing two lamps L_1 and L_2 is connected as shown.



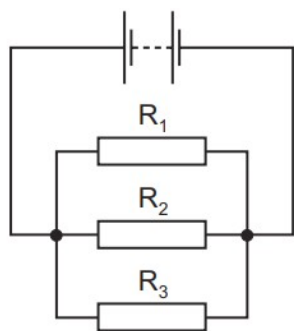
A voltmeter measures the potential difference across the lamp L_1 .

The filament of lamp L_1 breaks. What happens to the readings of the ammeter and of the voltmeter?

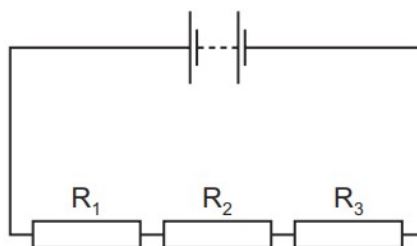
	reading on the ammeter	reading on the voltmeter
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

116. M/J/13/12/Q35

In the circuits P and Q below, resistors R_1 , R_2 and R_3 have different resistances.



circuit P



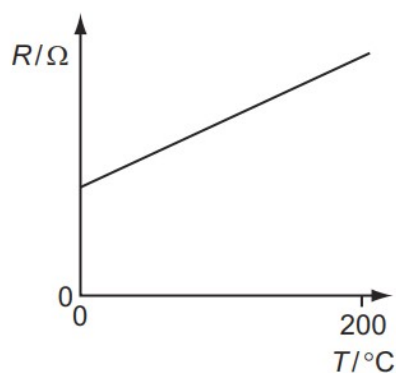
circuit Q

In which circuit are the currents in the resistors equal and in which circuit are the potential differences across the resistors equal?

	currents equal	potential differences equal
A	P	P
B	P	Q
C	Q	P
D	Q	Q

117. M/J/13/11/Q31

The electrical resistance of a metal wire varies with temperature as shown.



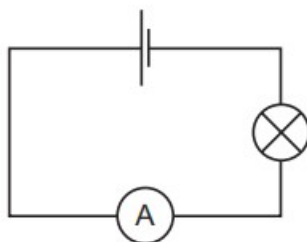
The resistance of the wire at 0°C is $1.00\ \Omega$ and at 100°C is $1.40\ \Omega$.

What is the resistance of the wire at 200°C ?

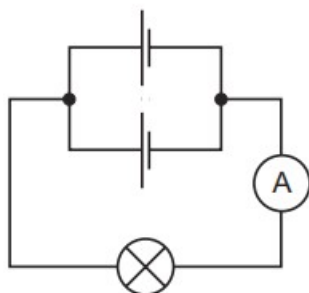
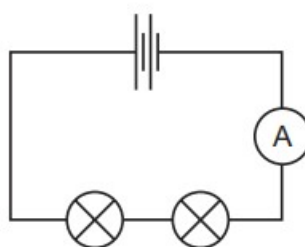
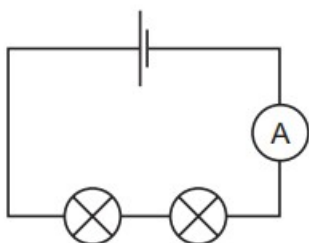
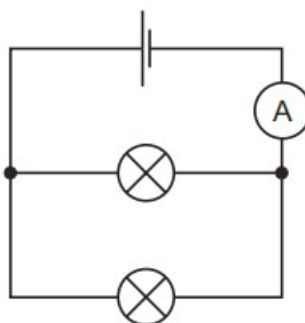
- A** $1.80\ \Omega$ **B** $1.96\ \Omega$ **C** $2.00\ \Omega$ **D** $2.80\ \Omega$

118. M/J/13/11/Q32

A cell is connected in series with an ammeter and a lamp. The reading on the ammeter is 1 A.



In which circuit, using identical lamps, ammeters and cells, is the reading on the ammeter 2 A?

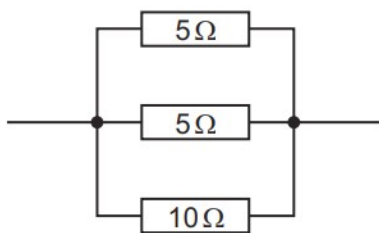
A**B****C****D****119. M/J/13/11/Q30**

Which of the following is equivalent to 1 V?

- A** 1 J/A
- B** 1 J/C
- C** 1 J/s
- D** 1 J/W

120. O/N/12/11/Q29

The diagram shows three resistors in parallel.

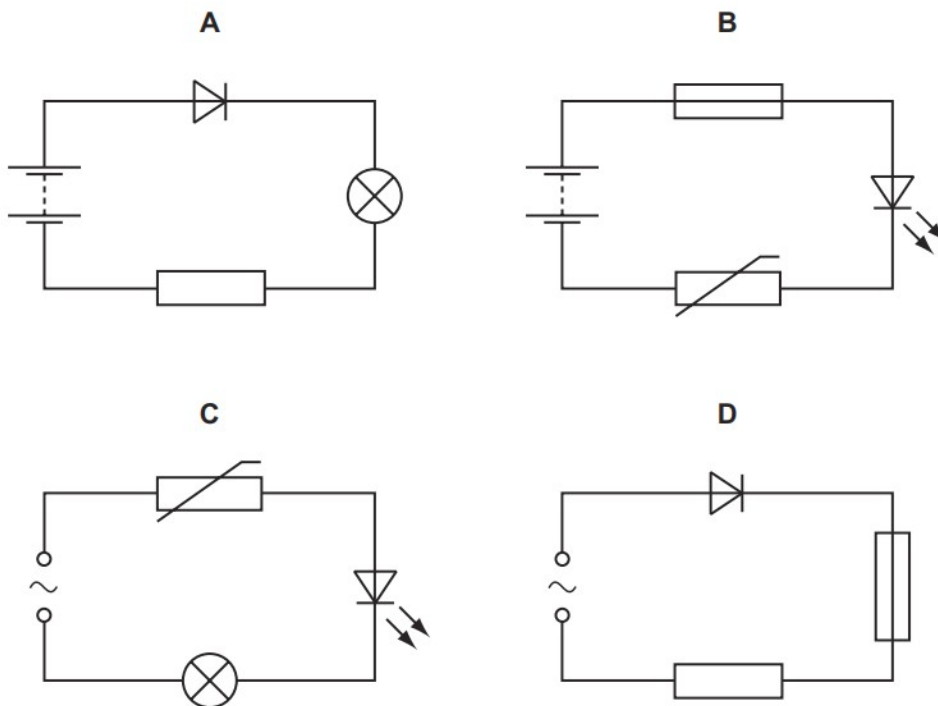


What is the combined resistance?

- A** $0.50\ \Omega$ **B** $2.0\ \Omega$ **C** $20\ \Omega$ **D** $250\ \Omega$

121. O/N/12/11/Q30

Which circuit contains a fuse and a rectifying diode?



122. O/N/12/12/Q28

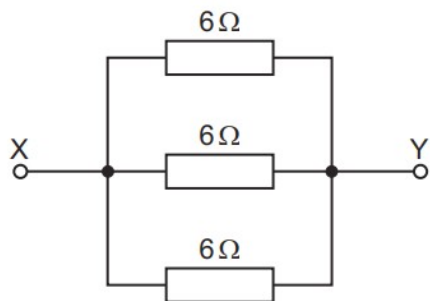
A metal wire has length l and cross-sectional area A .

What is the resistance proportional to?

- A** $A + l$ **B** $\frac{A}{l}$ **C** $\frac{l}{A}$ **D** $l \times A$

123. O/N/12/12/Q29

A student joins three 6Ω resistors as shown in the diagram.

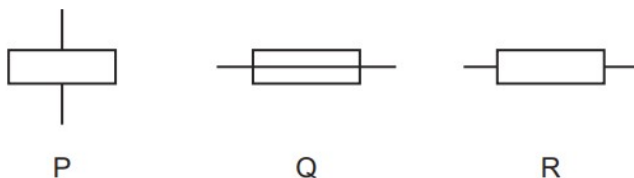


What is the total resistance between points X and Y?

- A** 0.5Ω **B** 2Ω **C** 6Ω **D** 18Ω

124. O/N/12/12/Q30

P, Q and R are electrical symbols.

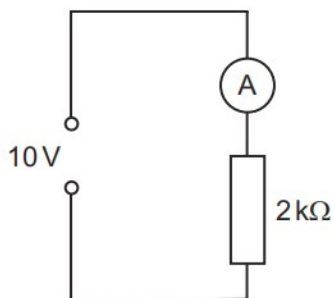


What do these symbols represent?

	P	Q	R
A	fuse	resistor	relay coil
B	relay coil	fuse	resistor
C	relay coil	resistor	fuse
D	resistor	fuse	relay coil

125. M/J/12/12/Q29

The diagram shows an ammeter connected in a circuit.

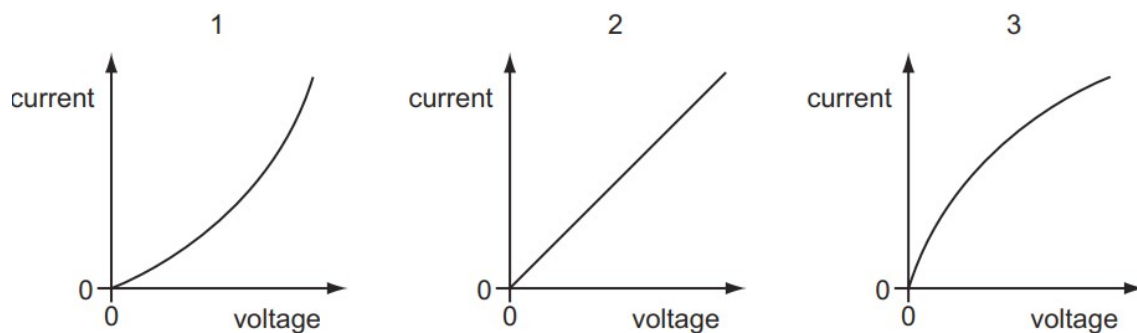


What is the current in the ammeter?

- A** 5 mA **B** 20 mA **C** 0.2 A **D** 5 A

126. M/J/12/12/Q28

The current/voltage graphs are for different electrical components.

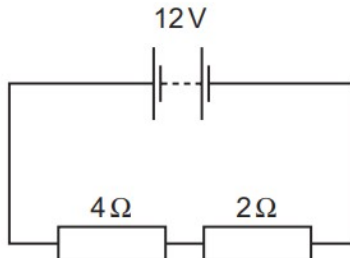


Which graph is for a resistor at constant temperature and which is for a filament lamp?

	resistor	lamp
A	1	2
B	2	1
C	2	3
D	3	2

127. M/J/12/12/Q30

In the circuit shown, the potential difference (p.d.) across the 4Ω resistor is 8 V .

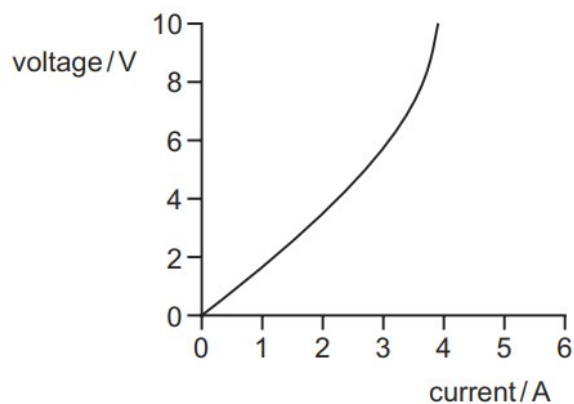


What is the p.d. across the 2Ω resistor?

- A** 4 V **B** 6 V **C** 8 V **D** 16 V

128. M/J/12/11/Q28

The voltage / current graph for a filament lamp is shown.

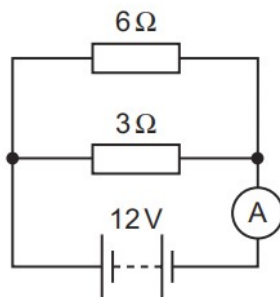


The voltage across the lamp increases. What happens to the temperature of the lamp and the resistance of the lamp?

	temperature	resistance
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

129. M/J/12/11/Q29

A 12 V battery is connected across a parallel arrangement of two resistors.



What is the reading on the ammeter?

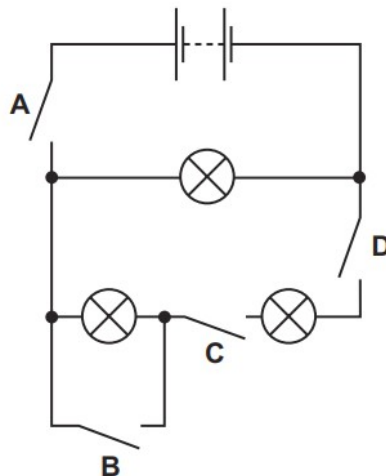
- A** 1.3 A **B** 2.0 A **C** 4.0 A **D** 6.0 A

130. M/J/12/11/Q30

The circuit shown is set up and then all the switches are closed.

One switch is now opened and all the lamps are lit.

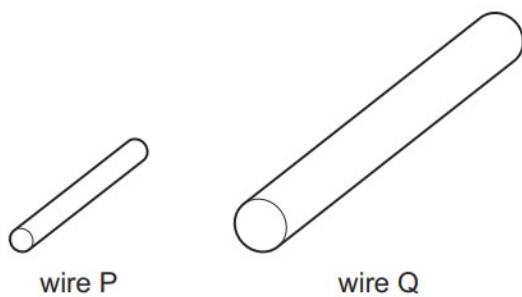
Which switch is opened?



131. O/N/11/11/Q28

The resistance of a cylindrical wire P is $80\ \Omega$. A second wire Q is made from the same material.

The cross-sectional area of Q is four times that of P. The length of Q is twice the length of P.

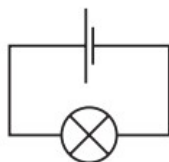


What is the resistance of Q?

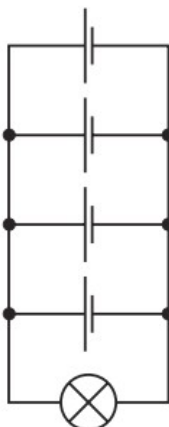
- A** 10Ω **B** 40Ω **C** 160Ω **D** 640Ω

132. O/N/11/11/Q29

A lamp, designed to work at 1.5 V, is connected to a cell of electromotive force (e.m.f.) 1.5 V. The lamp lights at normal brightness.



The lamp is now connected to four similar cells, each of e.m.f. 1.5 V, arranged in parallel.

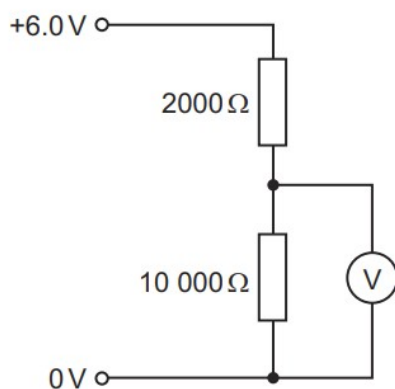


What is the effect of connecting the extra cells in this way?

- A** The lamp burns out.
- B** The lamp is dimmer.
- C** The lamp produces light for a longer time.
- D** The lamp produces light for a shorter time.

133. O/N/11/12/Q28

The diagram shows a potential divider system of two resistors connected to a 6.0 V power supply.

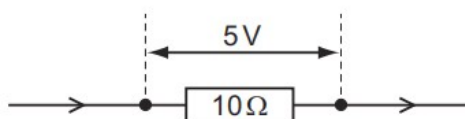


What is the reading on the voltmeter?

- A** 1.0 V
- B** 1.2 V
- C** 3.0 V
- D** 5.0 V

134. M/J/11/12/Q27

The potential difference (p.d.) across a 10Ω resistor is 5 V .



How much charge passes through the 10Ω resistor in 30 seconds?

- A** 2 C **B** 15 C **C** 60 C **D** 1500 C

135. M/J/11/12/Q28

Which changes both cause a decrease in the resistance of a copper wire?

	size of wire	temperature of wire
A	decrease in length	lower
B	increase in length	lower
C	decrease in thickness	higher
D	increase in thickness	higher

136. M/J/11/12/Q30

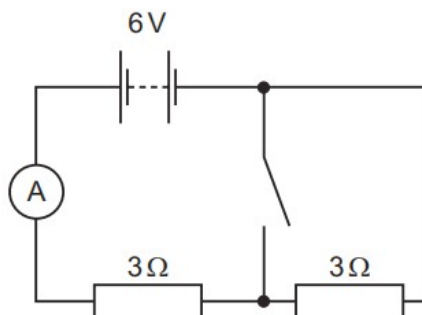
A lamp rated 6 V , 2 A is switched on for 60 s.

How much energy is used?

- A** 0.2 J **B** 20 J **C** 180 J **D** 720 J

137. M/J/11/12/Q31

The diagram shows a circuit.

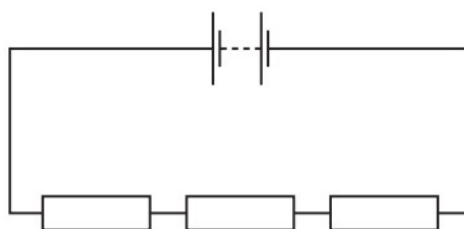


What is the reading on the ammeter when the switch is open, and the reading when it is closed?

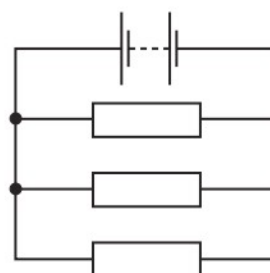
	ammeter reading when open / A	ammeter reading when closed / A
A	1	1
B	1	2
C	2	1
D	2	2

138. O/N/10/11/Q30

When three identical resistors are connected in series, their combined resistance is $6\ \Omega$.



series



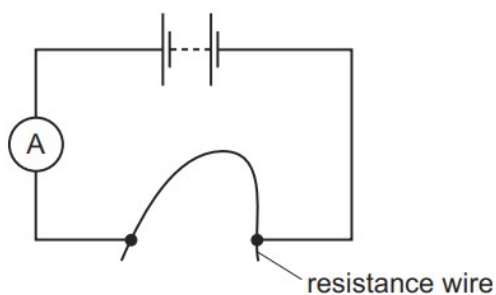
parallel

What is their combined resistance when they are connected in parallel?

- A** $\frac{1}{6}\ \Omega$ **B** $\frac{2}{3}\ \Omega$ **C** $\frac{3}{2}\ \Omega$ **D** $6\ \Omega$

139. M/J/10/12/Q27

A length of resistance wire is used as a resistor in a simple circuit.



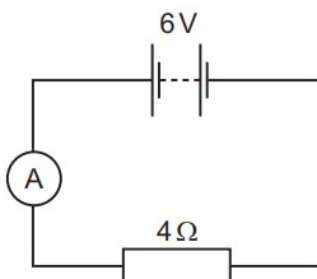
Four separate changes are made to the wire.

Which change will **not** reduce the value of the resistance of the wire?

- A** It is covered in an insulating sleeve.
B Its cross-sectional area is increased.
C Its length is decreased.
D Its temperature is decreased.

140. M/J/10/12/Q31

A 6 V supply is connected in series with an ammeter and a 4Ω resistor.



What is the reading on the ammeter?

- A** 0.67 A **B** 1.5 A **C** 10 A **D** 24 A

141. M/J/10/12/Q30

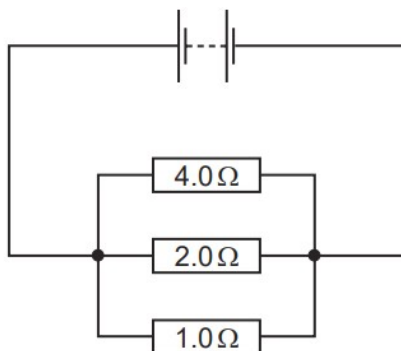
Sets of voltage-current readings are obtained for different electrical components.

Which set of readings is for a 100Ω resistor?

A	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-30	-15	-5	0	+5	+15	+30
B	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-30	-20	-10	0	+10	+20	+30
C	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-60	-40	-20	0	+20	+40	+60
D	voltage / V	-3	-2	-1	0	+1	+2	+3
	current / mA	-60	-45	-30	0	+30	+45	+60

142. O/N/09Q26

The circuit diagram shows three resistors in parallel with a battery.



What is the effective resistance of these three resistors?

- A** 0.57Ω **B** 0.86Ω **C** 1.75Ω **D** 7.00Ω

143. O/N/09/Q27

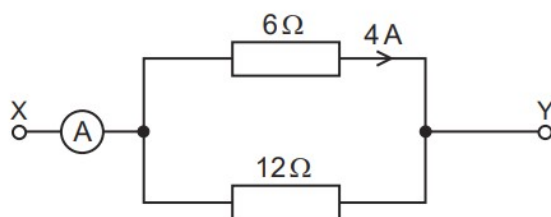
Ohm's law states that the current in a conductor is proportional to the potential difference across it, provided that a certain quantity remains constant.

What is this quantity?

- A length
- B pressure
- C temperature
- D thickness

144. O/N/09/Q28

Two resistors of $6\ \Omega$ and $12\ \Omega$ are arranged in parallel. A potential difference is connected across the terminals X and Y. The current in the $6\ \Omega$ resistor is $4\ \text{A}$.



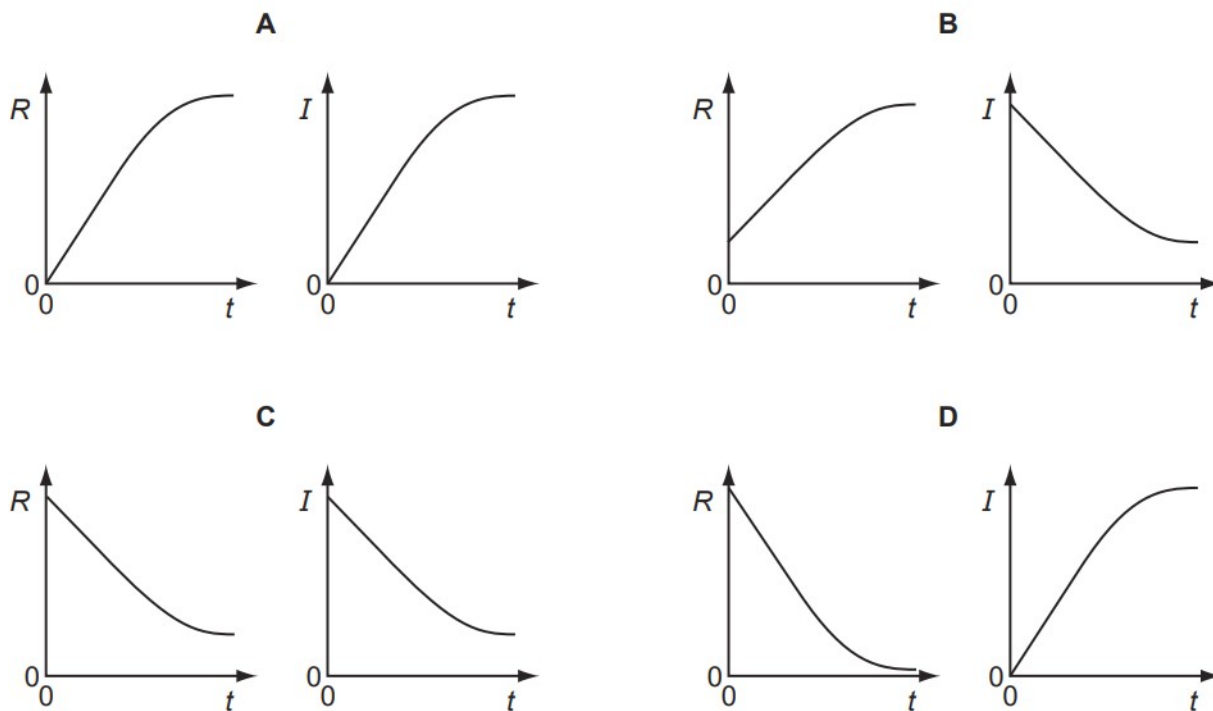
What is the current in the ammeter?

- A $4\ \text{A}$
- B $6\ \text{A}$
- C $8\ \text{A}$
- D $12\ \text{A}$

145. M/J/09/Q26

When a filament lamp is switched on, there is a current in the lamp. As the temperature of the filament rises, its resistance changes.

Which pair of graphs shows how the resistance R of the filament and the current I vary with time after the lamp is switched on?



146. O/N/08/Q27

A piece of polythene is rubbed with a cloth duster. The polythene becomes negatively charged and the cloth becomes positively charged.

What happens to the polythene and to the cloth to cause this?

	polythene	cloth
A	gains electrons	gains protons
B	gains electrons	loses electrons
C	loses protons	gains protons
D	loses protons	loses electrons

147. O/N/08/Q28

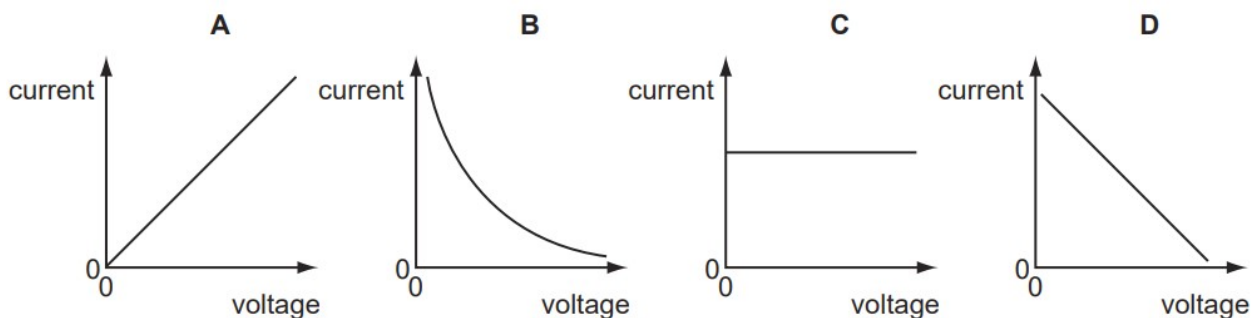
The current in an electric heater is 10A. It is switched on for five minutes.

How much charge flows through the heater?

- A** 0.5 C **B** 2 C **C** 50 C **D** 3000 C

148. O/N/08/Q29

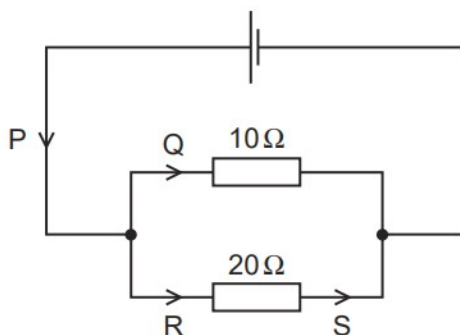
Which graph shows how the current changes when the voltage across a fixed resistance is varied?



149. O/N/08/Q30

The circuit diagram shows a parallel arrangement of resistors.

P, Q, R and S represent the current at the points shown.



Which statement is correct?

- A** P is greater than Q.
- B** Q is equal to R.
- C** R is greater than S.
- D** S is equal to P.

150. M/J/08/Q28

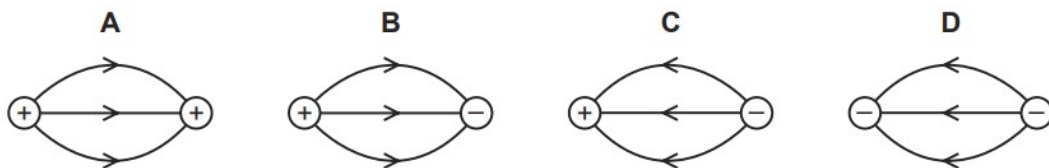
A perspex rod can be charged positively by rubbing it with a woollen cloth.

How does the rod gain its charge?

- A** The rod gains electrons.
- B** The rod gains protons.
- C** The rod loses electrons.
- D** The rod loses protons.

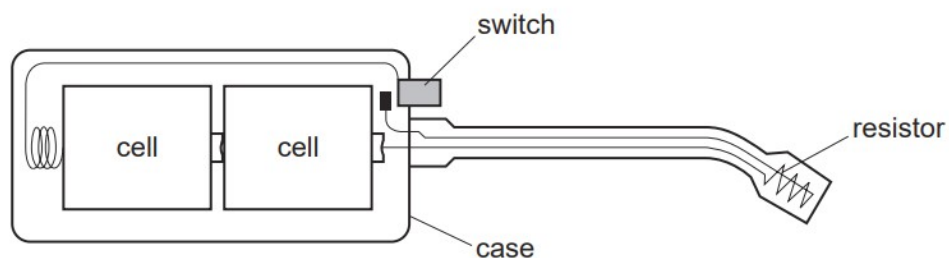
151. M/J/08/Q29

Which diagram correctly shows the electric field lines between two point charges?

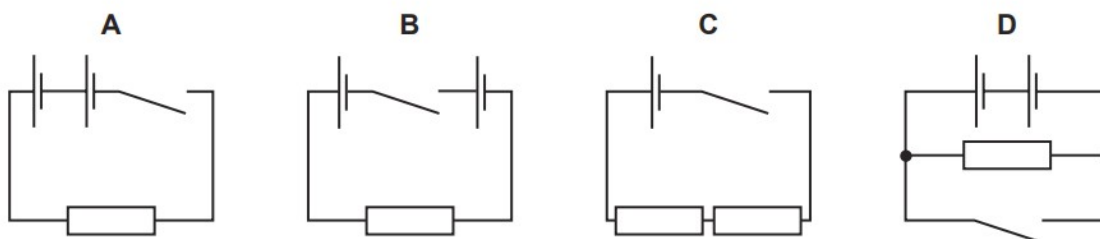


152. M/J/08/Q31

The diagram shows the components of a lighter for a gas cooker.



Which circuit diagram is correct for this lighter?



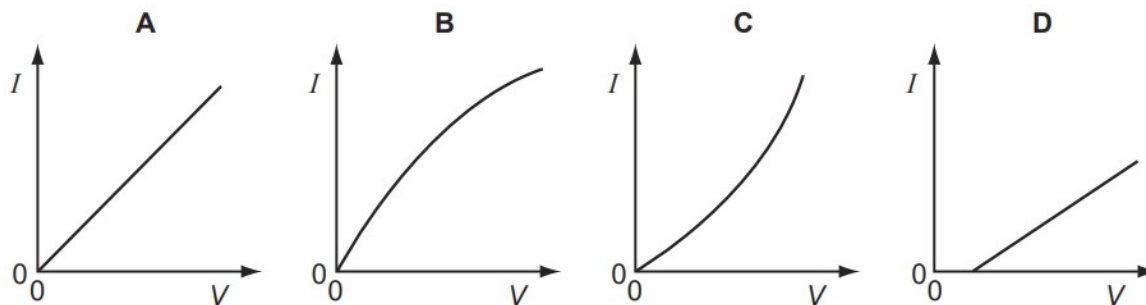
153. M/J/07/Q27

Which factors will both increase the resistance of a wire in a circuit?

	size of wire	temperature of wire
A	longer	lower
B	shorter	lower
C	thicker	higher
D	thinner	higher

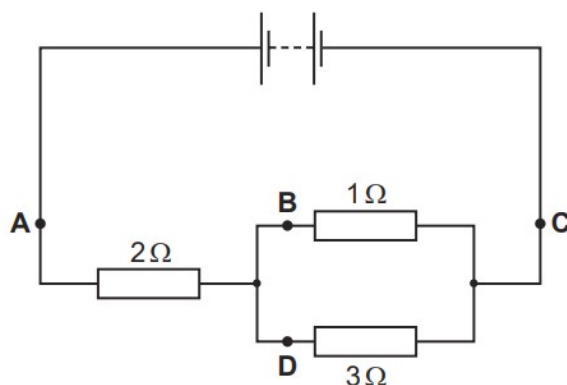
154. M/J/07/Q28

Which graph best represents how current I varies with voltage V in a component in which the resistance **increases** as the current **increases**.



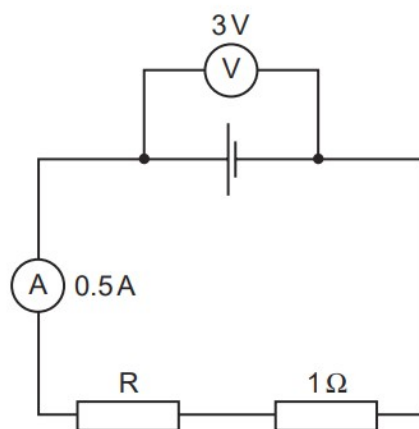
155. M/J/07/Q29

At which point in the circuit is the current the smallest?



156. M/J/07/Q30

The diagram shows a circuit.



The ammeter has negligible resistance.

What is the resistance of the resistor R?

- A** 0.5Ω **B** 1.5Ω **C** 5Ω **D** 6Ω

157. O/N/06/Q27

A wire has a resistance of 8Ω . A second wire, made of the same material, has half the length and twice the cross-sectional area.

What is the resistance of the second wire?

- A** 1Ω **B** 2Ω **C** 8Ω **D** 16Ω

158. M/J/06/Q30

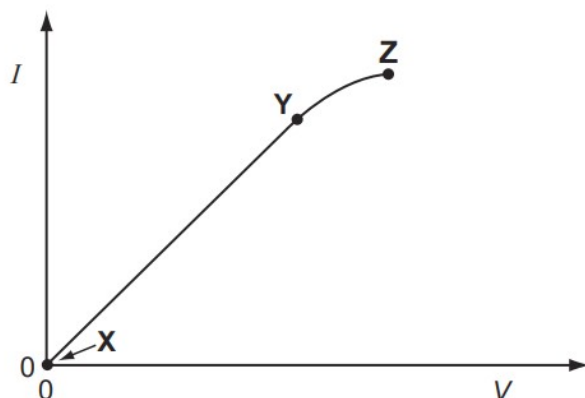
An electrical quantity is defined as 'the energy converted by a source in driving a unit charge round a complete circuit.'

What is this quantity called?

- A current
- B electromotive force
- C potential difference
- D power

159. M/J/06/Q31

The diagram shows the current I /voltage V graph for a length of resistance wire.



Where can Ohm's law be applied to the wire?

- A at Y only
- B at Z only
- C from X to Y
- D from X to Z

160. M/J/06/Q33

How much energy is converted in a resistor of 5.0Ω carrying a current of 2.0 A for 10 seconds?

- A 4.0 J
- B 25 J
- C 100 J
- D 200 J

161. M/J/06/Q32

Diagram 1 shows a resistor connected to a battery, an ammeter and a voltmeter.

The ammeter reading is 0.5 A and the voltmeter reading is 3.0 V .

A second identical resistor is now connected in parallel with the first resistor, as shown in diagram 2.

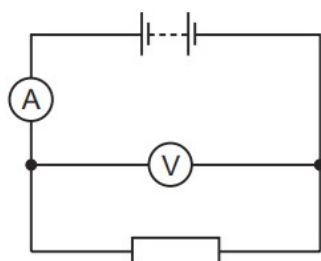


diagram 1

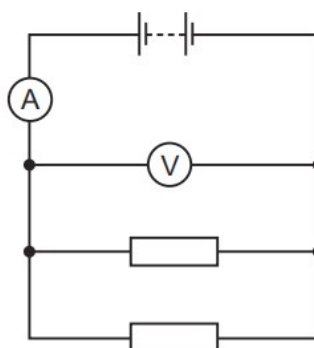


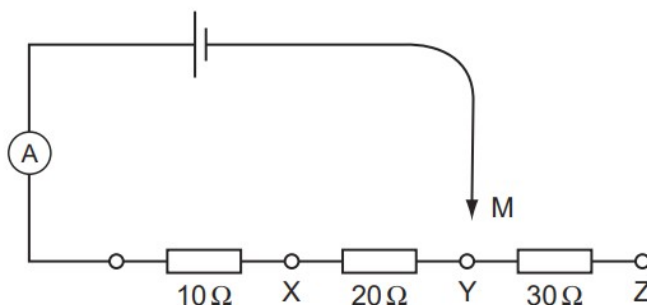
diagram 2

What are the ammeter and voltmeter readings in the circuit shown in diagram 2?

	ammeter reading / A	voltmeter reading / V
A	1.0	3.0
B	1.0	1.5
C	0.5	6.0
D	0.5	3.0

162. O/N/05/Q28

The diagram shows a cell connected in series with an ammeter and three resistors ($10\ \Omega$, $20\ \Omega$, $30\ \Omega$). The circuit can be completed by a moveable contact M.



When M is connected to X, the ammeter reads 0.6 A .

What is the ammeter reading when M is connected to Y?

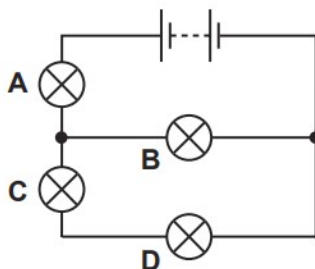
- A** 0.1 A **B** 0.2 A **C** 0.3 A **D** 0.6 A

163. O/N/05/Q29

In the circuit shown, the battery lights up all four lamps.

When one of the lamp filaments melts, the other three lamps stay on.

Which lamp filament melts?



164. M/J/05/Q27

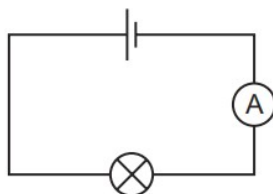
A cell is connected to a resistor.

What is the e.m.f. of the cell equal to?

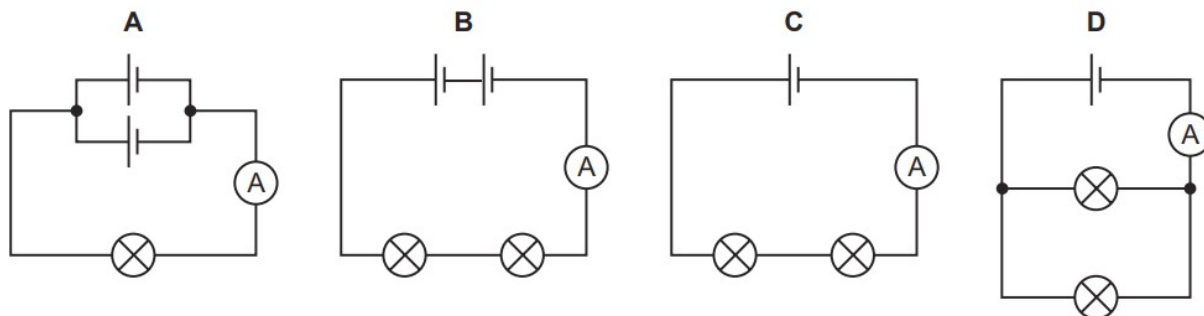
- A** The potential difference across the resistor for each unit of current.
- B** The power produced in the circuit for each unit of charge that passes.
- C** The work done in the circuit for each unit of charge that passes.
- D** The work done in the circuit for each unit of current.

165. M/J/05/Q28

A cell is connected in series with an ammeter and a lamp. The current is 1 A.

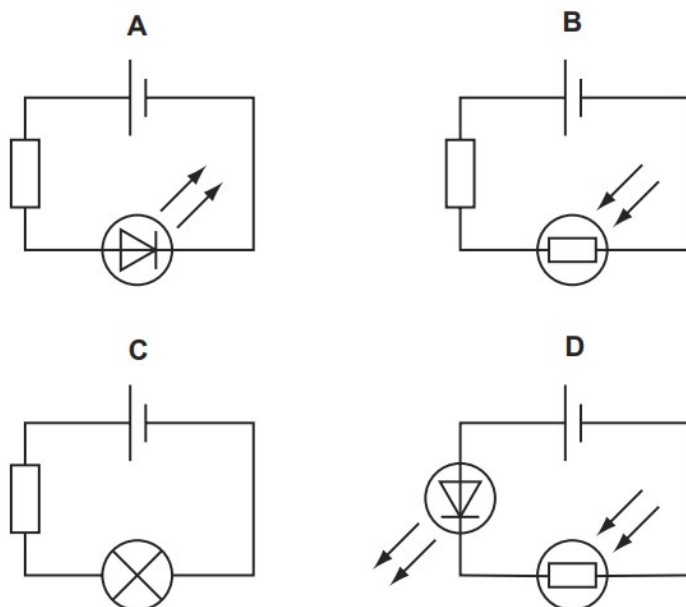


In which circuit, using identical cells, lamps and ammeters, is the current reading 2 A?



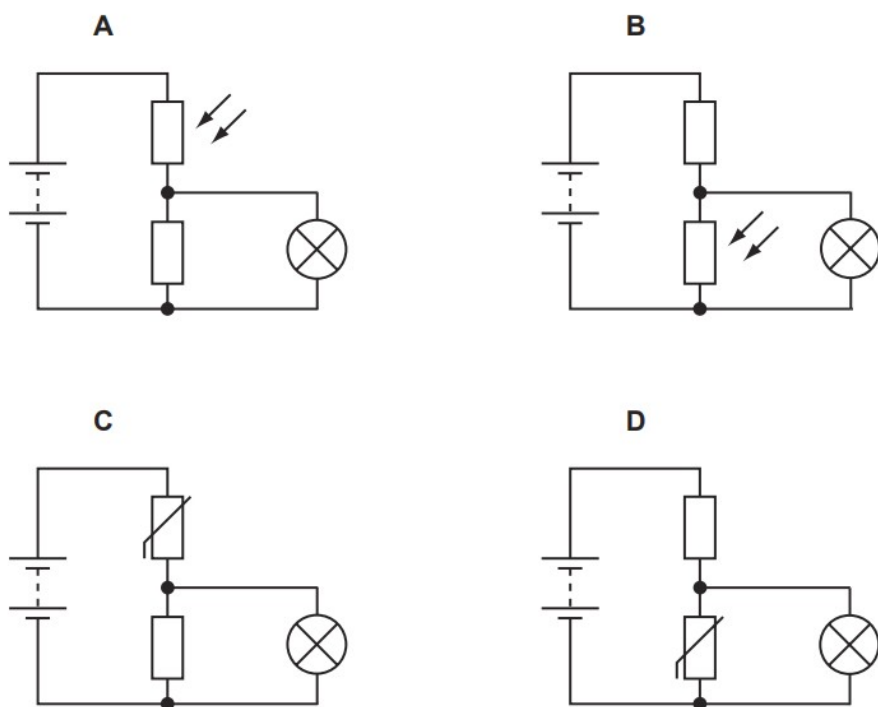
166. M/J/05/Q29

Which circuit contains a cell, a light-emitting diode and a fixed resistor?



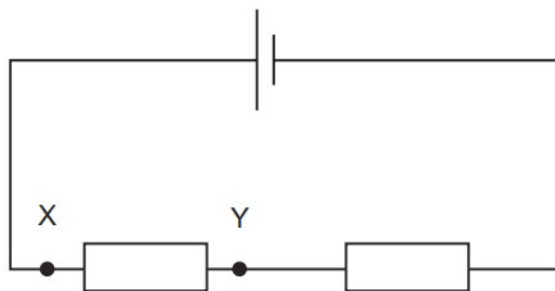
167. O/N/04/Q36

In which circuit will a lamp glow more brightly when less light shines on a light-dependent resistor?



168. M/J/04/Q28

The diagram shows two resistors connected in series with a cell.

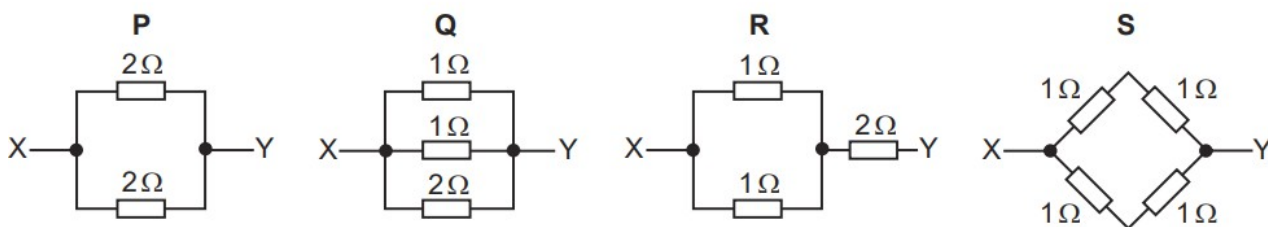


Which statement defines the potential difference across XY?

- A** the power needed to drive a unit charge through the cell
- B** the power needed to drive a unit charge between X and Y
- C** the work done in driving a unit charge through the cell
- D** the work done in driving a unit charge between X and Y

169. M/J/04/Q29

Which two resistor combinations have the same resistance between X and Y?



- A** P and Q
- B** P and S
- C** Q and R
- D** R and S